

Sovereign Risk with Endogenous Debt Limits*

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Abstract

Do sovereign debt ceilings matter if countries keep breaching or raising them? This paper proposes a new sovereign default model with long-term debt in which each government inherits a previously announced ceiling that it can breach by paying a cost and announces a new ceiling for the following period. This friction generates a state-dependent form of intermediate commitment. The ceiling mitigates debt dilution at the expense of fiscal flexibility. We show that ceilings that are backed by high breach costs lead to welfare gains. Conversely, when costs are low, impatient governments are tempted to gamble by promising tight ceilings that they then frequently breach. This can lead to welfare losses even relative to having no rule. Our contribution is to provide a positive theory of debt ceilings as endogenous, costly-to-breach commitment devices in sovereign debt markets.

Keywords: Debt Ceilings, Endogenous Commitment, Intermediate Commitment, Sovereign Default, Debt Dilution, Long-Term Debt.

JEL Classifications: E32, E44, F41, G01, G28.

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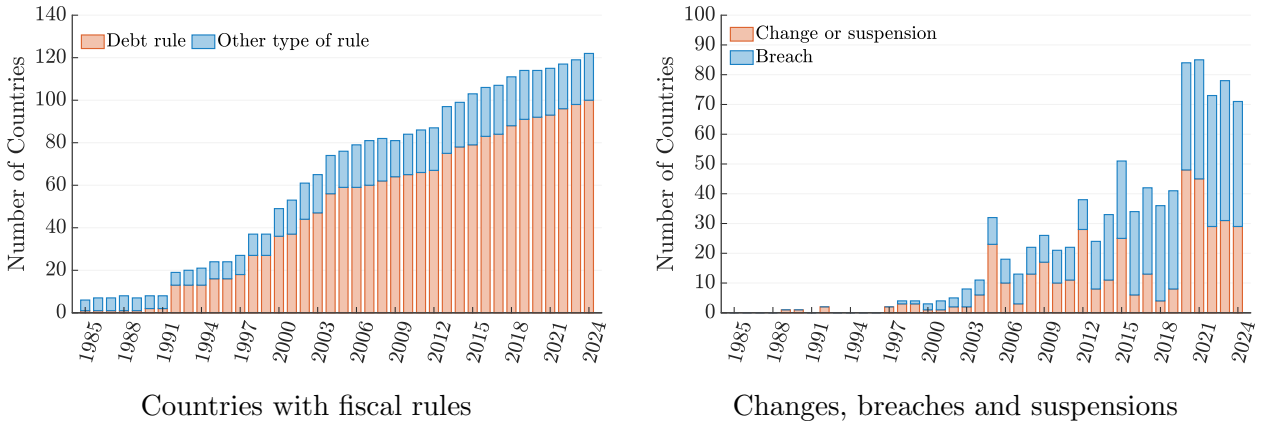
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1 Introduction

Sovereign bond markets price not only the fundamentals that determine a government’s capacity to repay, but also its incentives to dilute existing creditors by issuing additional debt. Such dilution incentives create a classic dynamic tension: governments borrow more in the future than is ex ante optimal, raising default risk and current borrowing costs. To manage these frictions, many countries have adopted fiscal rules—including debt ceilings, deficit limits, and expenditure caps—that function as policy anchors for fiscal behavior and have proliferated over the past four decades (left panel of Figure 1). Over the same period, changes, suspensions, and breaches of these rules have also become more frequent (right panel of Figure 1), suggesting that governments value the discipline fiscal rules provide but also face incentives to revise them when conditions deteriorate. Can governments adopt fiscal rules they may later overturn, and what are the implications for sovereign spreads and macroeconomic outcomes?

Figure 1: Trends in the Adoption and Revision of Fiscal Rules



Note: The figure documents the rise in fiscal rules and in subsequent changes, suspensions, and breaches. Data through 2021 come from the IMF Fiscal Rules Database. Data for 2022–2024 were compiled using IMF Article IV reports, the Inter-American Development Bank’s macroeconomic country reports, and country-specific sources. The updated series currently covers the 106 countries included in the 2021 IMF database.

In this paper, we consider an environment in which the government inherits a debt ceiling, decides if it wants to breach it, and chooses a debt ceiling for the future. The decision to breach a previously announced ceiling incurs a deviation cost. We interpret the deviation cost broadly as a reduced-form representation of political, procedural, reputational, or institutional frictions that make breaches of previously announced fiscal limits costly, while remaining

agnostic about their precise microfoundation.¹ In this environment, governments use ceilings as intermediate commitment devices that discipline future borrowing while balancing the loss of fiscal flexibility: ceilings that are too loose provide little commitment but are more likely to be respected, whereas ceilings that are too tight excessively constrain policy and are therefore more likely to be breached.

The underlying mechanism hinges on a simple trade-off that balances two economic forces. On the one hand, carrying the debt ceiling forward as a binding promise gives the government partial commitment not to issue excessive long-term bonds in the future, thereby mitigating the classic “debt dilution” bias in [Eaton and Gersovitz \(1981\)](#)-type models with long-term debt: when additional issuance today raises the probability of default tomorrow, investors anticipate the government’s ex post incentives to dilute legacy bondholders and demand higher yields, lowering bond prices ([Hatchondo and Martinez, 2009](#); [Chatterjee and Eyigungor, 2012](#)). On the other hand, the same ceiling restricts the government’s ability to respond to adverse shocks by issuing additional debt, which can make default more attractive in some states.²

We find that, in general, this partial-commitment mechanism reduces expected dilution and lowers average spreads and their volatility without materially reducing total borrowing. Interestingly, however, the welfare effects are non-monotone in the deviation cost. For low deviation costs, intermediate commitment can backfire and lead to welfare losses relative to no commitment in the absence of a rule. An impatient government—capturing short political horizons or political myopia—places relatively more weight on the current improvement in borrowing terms generated by a tight ceiling and less weight on the future costs incurred when that ceiling is breached. For low deviation costs, it may therefore be tempted to over-promise and under-deliver, choosing overly ambitious ceilings that are breached frequently. From the perspective of the more patient planner, the costs of these repeated breaches can outweigh the commitment gains. As deviation costs increase, breaches become less frequent and the gains from lower expected dilution dominate, raising welfare. Thus, in welfare terms, intermediate-commitment outcomes need not lie between the no-commitment economy without a rule and the commitment benchmark.

As suggested by the mechanism above, if self-imposed rules reduce dilution risk, they should lower sovereign spreads and their volatility. To provide simple motivating evidence

¹For example, violating a ceiling may lower the incumbent party’s probability of reelection. These costs are specific to breaching a domestic fiscal rule; sovereign default is disciplined separately by the economic costs of default.

²Because the value of flexibility varies with economic conditions, the resulting degree of commitment is itself state-dependent: a ceiling can discipline borrowing in normal times while being breached in adverse states. This is also a common rationale for escape clauses in fiscal rules.

for this prediction, we run the following cross-country regressions in the spirit of Cruces and Trebesch (2013):³

$$\text{EMBI Spread}_{i,t} = \alpha_i + \gamma_t + \beta \text{FiscalRule}_{i,t} + Z'_{i,t} \delta + \varepsilon_{i,t},$$

where the dependent variable is the EMBI Global sovereign spread (a monthly measure of hard-currency borrowing costs for emerging economies), $\text{FiscalRule}_{i,t}$ is a dummy equal to one when country i has an active fiscal rule at time t , and $Z_{i,t}$ collects the lagged restructuring and haircut variables, global risk proxies, and standard macro controls.⁴ Appendix Table 2 shows that the fiscal-rule coefficient β is consistently negative and statistically significant across several specifications, implying that countries with a fiscal rule exhibit EMBIG spreads roughly 40–60 basis points lower than otherwise similar economies. Moreover, Appendix Table 3 documents a parallel result for market volatility: fiscal-rule adopters display significantly lower monthly EMBIG volatility (a reduction of about 4–5 basis points in the standard deviation of daily returns). This empirical pattern aligns with the mechanism in our model, in which self-imposed fiscal constraints act as partial commitment devices that reduce perceived sovereign risk.

Our analysis proceeds in two steps. First, we use a three-period model to isolate the commitment problem and characterize how a debt ceiling changes the intertemporal allocation of borrowing. Under discretion, borrowing is tilted toward period 2 because the period-2 government does not internalize how additional issuance dilutes legacy bondholders. Anticipating this behavior, the period-1 government announces a ceiling on period-2 borrowing. When breaching the ceiling entails a fixed cost, the resulting allocation lies between the discretionary and full-commitment benchmarks. Once the breach cost exceeds a threshold that we characterize analytically, the ceiling makes the commitment allocation self-enforcing. Alternative breach-cost specifications show how the shape of the cost determines whether breaches occur in equilibrium and whether the allocation reaches the commitment benchmark at a finite threshold or only asymptotically.

We then introduce uncertainty into the three-period model to highlight the trade-off between commitment and fiscal flexibility. The government announces a single non-state-contingent ceiling before period-2 income is realized. For intermediate breach costs, govern-

³We use a sample of 57 countries up to 2019, augmenting their dataset with the fiscal-rule indicator constructed from the IMF Fiscal Rules Database (the same source underlying Figure 1). We extend the original Cruces–Trebesch dataset using data from Asonuma and Trebesch (2016) and Asonuma et al. (2023). For more details about the constructed series see Arce and Fourakis (2025).

⁴We interpret these regressions as motivating evidence only: fiscal-rule adoption is endogenous and may reflect the broader fiscal discipline that also lowers sovereign spreads.

ments pick a ceiling that can discipline borrowing following a favorable income realization but is breached following an adverse one, when the consumption-smoothing value of additional borrowing is greater. Greater income risk expands the range of breach costs over which this state-contingent response occurs. Thus, although the announced ceiling is non-state-contingent, the effective degree of commitment is state-dependent: the rule disciplines borrowing in normal times while preserving flexibility to handle adverse states.

Second, we develop a stochastic infinite-horizon model with long-term debt in which successive Markov governments make borrowing and default decisions and announce a non-state-contingent debt ceiling for the following period. The successor inherits this ceiling but may issue debt above it by paying a deviation cost. This setup adapts the partial-commitment mechanism of [Clymo, Lanteri and Villa \(2023\)](#) to sovereign debt limits. By embedding the announcement of future ceilings in a sovereign default environment with long-term debt, the model allows us to study how reducing dilution incentives affects bond prices. Self-imposed ceilings reduce average sovereign spreads and spread volatility, while average debt changes only modestly. The improvement in borrowing terms therefore comes primarily from limiting expected future dilution rather than from forcing the government to carry substantially less debt.

The welfare results highlight the non-monotonicity of intermediate commitment. For small breach costs, the rule provides a negligible amount of commitment while also imposing real costs when violations occur, leading to welfare that could be lower than in the no-commitment case; as the ceiling becomes more credible, breaches become rare, the benefits from lower expected dilution dominate, and welfare rises. We show that income risk directly affects the trade-off between commitment and fiscal flexibility. When income risk is lower, the government is tempted to choose ceilings that it ends up breaching frequently.

We also show that allowing the government to endogenously adjust the debt ceiling is quantitatively important. For each breach cost, we first compute the corresponding ergodic mean of the debt ceiling in the baseline economy and then solve a counterfactual in which the ceiling is held permanently at that value. Thus, the exercise holds fixed the average level of the ceiling and removes its state dependence. For low breach costs, the constant-ceiling model delivers higher welfare than both the baseline and the no-commitment case. In these cases, reducing the government's flexibility to adjust the ceiling prevents the government from choosing tight ceilings that it would breach. Conversely, for high breach costs, the ceilings imposed in the constant-ceiling model are too loose, and the model delivers virtually zero welfare gain compared to no commitment, while the baseline converges to the one-period-commitment welfare level. Thus, the value of state dependence depends on the credibility of

the rule: flexibility can be counterproductive when breach costs are low but important for realizing commitment gains when breach costs are high.

Finally, we show that self-imposed debt ceilings can limit the increase in spreads during crises, not primarily by reducing debt accumulation but by providing credibility to post-crisis fiscal consolidation.

Taken together, the results highlight a broader message for fiscal rules: the desirability of a rule is tied to how costly it is to violate it. More credible ceilings mitigate dilution and improve borrowing terms, but less credible ceilings can backfire because the commitment gains for impatient governments do not justify the frequent costs that breaching entails.

Related literature. This paper contributes to two strands of the literature: (i) sovereign default and fiscal rules, and (ii) partial commitment in optimal fiscal policy. The sovereign-default literature studies dilution and fiscal rules under exogenously given commitment frictions, while the partial-commitment literature in optimal fiscal policy develops costly-deviation mechanisms outside default environments. We bring the two together by developing a positive theory of debt-ceiling announcements as endogenous, partially enforceable commitment devices within a sovereign-default framework. The framework shows how governments can benefit from voluntarily restricting their successors’ choices, even when announced limits can be breached at a cost. These results also speak to the macro-finance literature by showing how endogenous fiscal rules can shape sovereign spreads through their effects on commitment, dilution risk, and credibility.

Sovereign Default and Fiscal Rules. In sovereign-debt models à la [Eaton and Gersovitz \(1981\)](#) with long-term debt, the “debt-dilution” time inconsistency highlighted by [Hatchondo and Martinez \(2009\)](#) and [Chatterjee and Eyigungor \(2012\)](#) generates persistent deficits: new bond issues dilute legacy claims, raise required yields, and lower bond prices.⁵ Subsequent work has quantified the welfare losses from dilution ([Aguiar et al., 2020](#)) and proposed both state-contingent rules that eliminate dilution ([Hatchondo et al., 2016](#)) and simpler rules that mitigate its impact ([Hatchondo et al. 2022a](#); [Roch and Roldán 2023](#)), evaluating gains by comparing commitment versus discretionary equilibria. Relatedly, [Mateos-Planas et al. \(2025\)](#) examine how different exogenously imposed forms of commitment—such as commitment to default thresholds or to continuation prices—alter default outcomes in standard sovereign-debt models. Their commitment structures are taken as given and operate directly on

⁵It should be noted that fiscal rules are not the only way to handle dilution. [Dovis \(2019\)](#), for instance, shows that lenders could adopt history-dependent bond pricing mechanisms that would make the government internalize the dilution problem.

default conditions. In our framework, commitment instead arises endogenously: governments choose debt-ceiling announcements period by period to constrain their successors, who may breach them only by incurring a cost. This mechanism generates state-dependent degrees of partial commitment as an equilibrium outcome and shows how governments can benefit from voluntarily restricting the discretion of their future selves. It thereby complements the normative “rules versus flexibility” literature initiated by [Amador et al. \(2006\)](#) and developed in [Halac and Yared \(2014, 2017, 2020, 2022\)](#) by showing how fiscal rules operate when the government is endowed with an endogenous, state-dependent choice of rule—a debt ceiling—that it strategically announces each period and uses in equilibrium to discipline its future selves, while allowing successors to breach it at a political and institutional cost.⁶

Optimal Fiscal Policy with Partial Commitment and Fiscal Announcement. Our framework is related to the literature on optimal fiscal policy under limited commitment and on fiscal announcements. First, on intermediate commitment (“partial commitment”), we build on the idea that governments make non-contingent announcements but face costs when deviating from them. Papers in this literature that study partial commitment typically model re-optimization as arriving exogenously (e.g. [Debortoli and Nunes, 2010, 2013](#)), treating deviations as opportunities that arise independently of the state. Unlike these approaches, we adopt the costly-deviation mechanism of [Clymo et al. \(2023\)](#) and adapt it to sovereign debt ceilings and default, so that the extent of reneging—and thus the degree of commitment—is determined endogenously by the model’s state variables. This state-dependent slack then shapes strategic debt-ceiling announcements. Relatedly, [Farhi \(2010\)](#), [Klein et al. \(2008\)](#), and [Karantounias \(2019\)](#) use generalized Euler–equation methods to explore time consistency and default, while [Clymo and Lanteri \(2020\)](#) show that even short-horizon commitment can sustain first-best outcomes. We extend their approach by introducing costly, state-contingent reneging of debt ceilings in a stochastic economy with sovereign default, where dilution incentives, endogenous default risk, and bond pricing jointly determine the value of commitment. A related mechanism is studied by [Bocola et al. \(2026\)](#) in the context of the credibility of central-bank mandates under fiscal–monetary interactions.

Second, on fiscal announcements per se, we bridge optimal-policy theory with the empirical and quantitative work that treats announcements as exogenous drivers of expectations. Empirical papers such as [Mertens and Ravn \(2012\)](#) and [Alesina et al. \(2015\)](#) document the macro effects of announced plans, and quantitative studies like [Mertens and Ravn \(2011\)](#) and [Fernández-Villaverde et al. \(2015\)](#) embed announcement “shocks” in DSGE settings.

⁶See also [Espino et al. \(2022\)](#) for an analysis of fiscal-rule suspensions during the COVID-19 crisis.

By distinguishing between announced and implemented policies—and by endogenizing the cost of deviating from announcements—we embed insights from the empirical literature on fiscal announcements in an optimal-policy framework. Our approach also relates to the fiscal-rules literature (e.g., King et al. 1988; Schmitt-Grohe and Uribe 1997; Athey et al. 2005; Halac and Yared 2014), which shows that limits on state contingency can amplify fluctuations. We demonstrate that costly, partial state contingency—driven by political constraints on commitment and strategic debt-ceiling announcements in a sovereign-default environment—generates rich dynamics with meaningful policy implications for the design of fiscal rules and debt-management frameworks.

Structure of the paper. The remainder of the paper is organized as follows. Section 2 presents a three-period model and derives analytical results on debt dilution, intermediate commitment, and state-contingent ceiling breaches. Section 3 develops the stochastic infinite-horizon model with endogenous debt-ceiling announcements and quantifies their implications for sovereign spreads, borrowing, breaches, welfare, and the value of state dependence. Section 4 concludes.

2 Three-Period Model

This section presents a tractable three-period model that isolates the core mechanism behind our debt-ceiling results.⁷ Subsection 2.1 characterizes how a debt ceiling can sustain allocations between the discretionary and full-commitment benchmarks, while Subsection 2.2 introduces uncertainty to show how the same ceiling can balance commitment in normal times with flexibility in adverse states.

Environment. Time is discrete with three periods, $t \in \{1, 2, 3\}$. Output is zero in the first two periods, $y_1 = y_2 = 0$, and equals $\bar{y} > 0$ in the final period. The government borrows from risk-neutral foreign investors in periods 1 and 2 and has preferences

$$\sum_{t=1}^3 \beta^{t-1} u(c_t), \quad u(c) = -\frac{1}{c}.$$

⁷The setup is similar to the three-period model in Hatchondo et al. (2022b), but extends their framework by introducing the concept of intermediate commitment.

Debt issued in periods 1 and 2 matures in period 3. Let b_1 denote debt issued at $t = 1$, b_2 debt issued at $t = 2$, and

$$B \equiv b_1 + b_2$$

total obligations due at maturity. Define repayment slack as

$$x \equiv \frac{\bar{y} - B}{\bar{y}} \in (0, 1], \quad B = \bar{y}(1 - x). \quad (1)$$

Thus x measures fiscal space at repayment: lower x corresponds to higher debt and a lower capacity to repay.

Default, pricing, and terminal utility. At $t = 3$, the government chooses whether to repay or default. If it repays, it consumes $\bar{y} - B = \bar{y}x$. If it defaults, it consumes $\bar{y}/\theta$, where θ is a random default-cost parameter distributed Pareto on $[1, \infty)$ with shape parameter $\alpha = \frac{1}{2}$. Default occurs whenever default consumption exceeds repayment consumption:

$$\frac{\bar{y}}{\theta} > \bar{y} - B \quad \Longleftrightarrow \quad \theta < \frac{1}{x}.$$

With risk-neutral lenders and a unit gross risk-free rate, the bond price equals the repayment probability:

$$q(B) = \Pr(\theta > 1/x) = x\alpha = \sqrt{x}. \quad (2)$$

Hence higher debt lowers fiscal space, raises default risk, and reduces the bond price. Given the default rule, terminal consumption is

$$c_3 = \begin{cases} \bar{y} - B, & \text{if the government repays,} \\ \bar{y}/\theta, & \text{if the government defaults.} \end{cases} \quad (3)$$

Using the Pareto distribution and $u(c) = -1/c$, expected terminal utility is

$$\mathbb{E}[u(c_3) \mid B] = \frac{1}{\bar{y}} (1 - 2x^{-1/2}). \quad (4)$$

Expression (4) summarizes how higher debt lowers welfare by reducing the probability of repayment and worsening consumption in repayment states. The derivation is reported in Appendix A.

2.1 Sustaining Commitment through a Debt Ceiling

We use the three-period framework to compare three allocations: full commitment, no commitment, and intermediate commitment through a debt ceiling. The commitment and no-commitment benchmarks isolate the dynamic inconsistency: the period-2 government overborrows because it does not internalize how new issuance dilutes legacy creditors and raises default risk. We then introduce a ceiling $\bar{b}$ chosen by the period-1 government. If the period-2 government borrows above the inherited ceiling, it pays

$$\Phi(b_2, \bar{b}) = \phi(b_2 - \bar{b})^\zeta \mathbf{1}_{\{b_2 > \bar{b}\}}, \quad \phi \geq 0, \zeta \geq 0. \quad (5)$$

The parameter ϕ governs enforcement strength, while ζ governs the shape of breach costs. We focus on three cases: a fixed cost ($\zeta = 0$), a piecewise-linear cost ($\zeta = 1$), and a quadratic cost ($\zeta = 2$). In all cases, a positive breach cost disciplines period-2 borrowing and allows the period-1 government to move the allocation toward the commitment benchmark.

Commitment. Under full commitment, the period-1 government chooses both b_1 and b_2 , thereby fixing the entire borrowing path and internalizing how total obligations affect bond prices and default risk. The commitment problem is

$$V_1^C = \max_{b_1, b_2} \{u(c_1) + \beta u(c_2) + \beta^2 \mathbb{E}[u(c_3) \mid B]\}, \quad c_t = q(B)b_t, \quad B = b_1 + b_2.$$

Let (b_1^C, b_2^C) denote the solution. The first-order conditions are

$$u'(c_1)(q(B) + b_1 q'(B)) + \beta u'(c_2)b_2 q'(B) + \beta^2 \frac{\partial}{\partial B} \mathbb{E}[u(c_3) \mid B] = 0, \quad (6)$$

$$u'(c_1)b_1 q'(B) + \beta u'(c_2)(q(B) + b_2 q'(B)) + \beta^2 \frac{\partial}{\partial B} \mathbb{E}[u(c_3) \mid B] = 0. \quad (7)$$

These conditions differ only in which issuance margin receives the direct current-resource benefit. Combining them yields a standard Euler equation

$$\beta \frac{u'(c_2)}{u'(c_1)} = 1, \quad (8)$$

and, using $u(c) = -1/c$,

$$b_2^C = \sqrt{\beta} b_1^C.$$

Thus the commitment allocation equalizes discounted marginal utilities across periods 1 and 2 and tilts borrowing toward period 1. The key feature is that the planner internalizes the

effect of each issuance decision on total debt, repayment risk, and the common bond price $q(B)$; this is precisely the margin that will be distorted under no commitment.

Discretion. Without commitment, decisions are sequential. In period 2, the government takes the inherited debt stock b_1 as given and chooses b_2 to maximize

$$V_2^{NC}(b_1) = \max_{b_2} \{u(c_2) + \beta \mathbb{E}[u(c_3) \mid B]\}, \quad c_2 = q(B)b_2, \quad B = b_1 + b_2.$$

The first-order condition is

$$u'(c_2)(q(B) + b_2 q'(B)) = -\beta \frac{\partial}{\partial B} \mathbb{E}[u(c_3) \mid B]. \quad (9)$$

Solving (9) yields the period-2 best response

$$b_2^\dagger(b_1) = \frac{\bar{y}}{4\beta} \left[-3 + \sqrt{9 + 16\beta \left(1 - \frac{b_1}{\bar{y}} \right)} \right]. \quad (10)$$

The policy $b_2^\dagger(b_1)$ is decreasing in inherited debt: a larger b_1 raises total obligations, lowers the bond price, and makes additional period-2 borrowing less attractive.

At time 1, the government and lenders anticipate the reoptimization rule $b_2^\dagger(b_1)$. Hence the period-1 government chooses b_1 taking into account both the direct effect of b_1 on total debt and the induced response of the successor's borrowing decision. In what follows, total debt is evaluated along the period-2 best response, so that $B(b_1) = b_1 + b_2^\dagger(b_1)$. Combining the period-2 condition (9) with the period-1 optimality condition yields the generalized Euler equation⁸

$$\beta \frac{u'(c_2)}{u'(c_1)} = 1 + b_1 \frac{q'(B(b_1))}{q(B(b_1))} (1 + b_2^{\dagger'}(b_1)). \quad (11)$$

Relative to the commitment condition (8), the additional term in blue captures the strategic response of period-2 borrowing to inherited debt. Since $q'(B) < 0$, this term distorts the intertemporal Euler equation away from the commitment benchmark: the period-1 government anticipates that its successor will reoptimize, while the period-2 government does not internalize the effect of new issuance on legacy creditors. This strategic distortion generates excessive short-term borrowing relative to commitment.

⁸See Appendix A.2 for details.

Excessive Period-2 Borrowing Under No Commitment

Proposition 1. Fix $\beta \in (0, 1)$ and $\bar{y} > 0$. Let (b_1^C, b_2^C) be the commitment allocation and let (b_1^{NC}, b_2^{NC}) be the no-commitment allocation. Then borrowing undertaken at $t = 2$, as a share of total borrowing, is weakly higher under discretion:

$$\frac{b_2^{NC}}{b_1^{NC} + b_2^{NC}} \geq \frac{b_2^C}{b_1^C + b_2^C} = \frac{\sqrt{\beta}}{1 + \sqrt{\beta}}.$$

Proof. See Appendix A.2.

Proposition 1 formalizes the overborrowing motive. In the discretionary economy, borrowing is tilted toward period 2 because the successor does not internalize how additional issuance dilutes existing claims and raises default risk.

Intermediate commitment with a fixed cost ($\zeta = 0$). We now introduce a borrowing ceiling $\bar{b}$ for period-2 issuance. If the period-2 government borrows above the inherited ceiling, it pays a fixed cost $\phi \geq 0$. Its problem is

$$V_2(b_1, \bar{b}) = \max_{b_2} \{u(c_2) - \phi \mathbf{1}_{\{b_2 > \bar{b}\}} + \beta \mathbb{E}[u(c_3) \mid B]\}, \quad c_2 = q(B)b_2, \quad B = b_1 + b_2.$$

Let

$$f_0(b_2 \mid b_1) \equiv u(c_2) + \beta \mathbb{E}[u(c_3) \mid B]$$

denote the period-2 payoff excluding the ceiling cost. Given $(b_1, \bar{b})$, the successor compares respecting the ceiling with paying the cost and choosing the unconstrained discretionary policy $b_2^\dagger(b_1)$:

$$b_2(b_1, \bar{b}) = \begin{cases} b_2^\dagger(b_1), & \text{if } f_0(b_2^\dagger(b_1) \mid b_1) - \phi > f_0(\bar{b} \mid b_1), \\ \bar{b}, & \text{otherwise.} \end{cases} \quad (12)$$

Thus a fixed-cost ceiling generates a simple incentive constraint. For a given b_1 , the ceiling is enforceable if the successor weakly prefers respecting it:

$$f_0(b_2^\dagger(b_1) \mid b_1) - \phi \leq f_0(\bar{b} \mid b_1).$$

Among enforceable ceilings, the period-1 government chooses the tightest one, because tighter ceilings provide greater discipline as long as the successor is willing to respect them. Hence, when the commitment allocation is not yet self-enforcing, the optimal ceiling lies on the

indifference locus

$$f_0(b_2^\dagger(b_1) \mid b_1) - \phi = f_0(\bar{b} \mid b_1). \quad (13)$$

At this ceiling, the successor is indifferent between deviating to $b_2^\dagger(b_1)$ and respecting $\bar{b}$, and equilibrium borrowing satisfies $b_2(b_1, \bar{b}) = \bar{b}$.

The smallest fixed cost that makes the commitment allocation self-enforcing is the value gap between the successor's unconstrained deviation and the commitment choice:

$$\phi_{\min} = f_0(b_2^\dagger(b_1^C) \mid b_1^C) - f_0(b_2^C \mid b_1^C) > 0. \quad (14)$$

For $\phi \geq \phi_{\min}$, setting $\bar{b} = b_2^C$ sustains the commitment allocation. For $0 < \phi < \phi_{\min}$, the ceiling partially disciplines period-2 borrowing, and the equilibrium lies between the no-commitment and commitment benchmarks.

To characterize the period-1 allocation in the intermediate region, use the binding incentive constraint (13) to implicitly express the chosen ceiling as a function of b_1 , denoted $\bar{b}(b_1)$. Since the successor then respects the ceiling, equilibrium period-2 borrowing satisfies $b_2 = \bar{b}(b_1)$, and total debt along this constraint locus is $B(b_1) = b_1 + \bar{b}(b_1)$. The period-1 generalized Euler equation is

$$\begin{aligned} & u'(c_1)(q(B(b_1)) + b_1 q'(B(b_1))(1 + \bar{b}'(b_1))) \\ & + \beta \left[u'(c_2)(q(B(b_1))\bar{b}'(b_1) + q'(B(b_1))\bar{b}(b_1)(1 + \bar{b}'(b_1))) + \beta \mathcal{U}'_3(B(b_1))(1 + \bar{b}'(b_1)) \right] = 0, \end{aligned} \quad (15)$$

where $\mathcal{U}_3(B) \equiv \mathbb{E}[u(c_3) \mid B]$. This equation has the same structure as the discretionary generalized Euler equation, but the successor's unconstrained response $b_2^{\dagger'}(b_1)$ is replaced by the ceiling response $\bar{b}'(b_1)$. The debt ceiling therefore changes the strategic policy-response term that distorts the intertemporal allocation. Detailed derivations are provided in Appendix A.4.

Putting these objects together yields the threshold characterization in Proposition 2. Intuitively, the fixed-cost ceiling mitigates the period-2 overborrowing motive. Low costs provide little discipline; intermediate costs partially restrict the successor and generate allocations between no commitment and full commitment; and costs above $\phi_{\min}$ make the commitment allocation self-enforcing, so additional increases in ϕ are redundant.

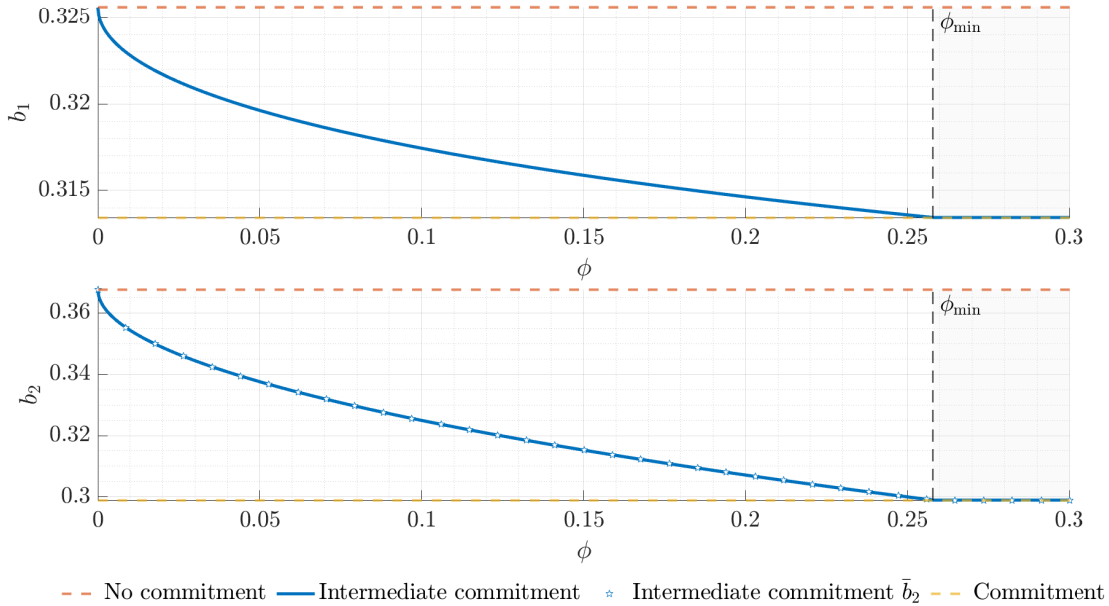
Sustainable Allocations Under a Debt Ceiling

Proposition 2. Fix a cost $\phi \geq 0$. The solution to the intermediate-commitment problem is a pair (b_1^*, b_2^*) such that:

1. If $\phi = 0$, the unique equilibrium is the no-commitment allocation $(b_1^*, b_2^*) = (b_1^{NC}, b_2^{NC})$, given by (10) and (9).
2. If $0 < \phi < \phi_{\min}$, the equilibrium is an intermediate-commitment allocation (b_1^*, b_2^*) characterized by (13) and (15), with $b_1^* \in (b_1^C, b_1^{NC})$ and $b_2^* \in (b_2^C, b_2^{NC})$. The planner sets $\bar{b} = b_2^*$.
3. If $\phi \geq \phi_{\min}$, the planner sets $\bar{b} = b_2^C$, and the economy sustains the commitment allocation $(b_1^*, b_2^*) = (b_1^C, b_2^C)$, given by (6) and (7), as a self-enforcing outcome at $t = 2$.

Proof. See Appendix A.4.

Figure 2: Optimal Debt Positions Under Varying Commitment Strength



Note. The figure plots optimal debt choices b_1 and b_2 , as functions of the cost ϕ . The blue solid line represents the case of intermediate commitment, while the red and yellow dashed lines correspond to the no-commitment and full-commitment benchmarks, respectively. The blue star markers plot the endogenous ceiling. The shaded gray region marks values of ϕ above the minimum threshold $\phi_{\min}$ required to sustain commitment.

Figure 2 further illustrates this threshold logic. For $\phi = 0$, the economy coincides with the

no-commitment benchmark. As ϕ rises, the ceiling tightens and period-2 borrowing moves toward the commitment level. Once ϕ reaches $\phi_{\min}$, the commitment allocation is sustained, and further increases in the breach cost have no additional effect.

Intermediate commitment with a piecewise-linear cost ($\zeta = 1$). When $\zeta = 1$, borrowing above the ceiling entails the cost

$$\Phi(b_2, \bar{b}) = \phi(b_2 - \bar{b})\mathbf{1}_{\{b_2 > \bar{b}\}}.$$

The objective is continuous but kinked at $b_2 = \bar{b}$. Below the ceiling, the penalty is inactive, so the borrowing condition is the same as in the no-commitment problem. Above the ceiling, an interior breaching choice satisfies

$$\frac{\partial}{\partial b_2} f_0(b_2 \mid b_1) = \phi, \quad b_2 > \bar{b},$$

so the linear cost introduces a constant marginal wedge in the successor's borrowing condition.

In this case, Figure 3 shows that anticipated breaches are not optimal. To see why, suppose period 1 chooses a ceiling $\bar{b}$ and anticipates that period 2 will choose $b_2 > \bar{b}$. Raising the ceiling slightly, while keeping it below the anticipated b_2 , does not change the successor's marginal condition: the marginal cost of borrowing above the ceiling remains ϕ . Hence the implemented borrowing choice is unchanged. But the total cost paid by the successor falls because the violation $b_2 - \bar{b}$ is smaller. The period-1 government therefore prefers to raise the ceiling until it coincides with the implemented borrowing level. As a result, when the linear-cost ceiling is active in the deterministic model, the optimal endogenous ceiling binds:

$$b_2 = \bar{b}.$$

Figure 3 illustrates this property. As ϕ increases, the linear cost progressively disciplines period-2 borrowing, and the allocation reaches the commitment benchmark at a finite threshold.

Intermediate commitment with a quadratic cost ($\zeta = 2$). The quadratic specification provides a useful contrast to the fixed- and linear-cost cases because violations are smooth rather than discrete. Given $(b_1, \bar{b})$, the period-2 government solves

$$V_2(b_1, \bar{b}) = \max_{b_2} \left\{ f_0(b_2 \mid b_1) - \phi(b_2 - \bar{b})^2 \mathbf{1}_{\{b_2 > \bar{b}\}} \right\},$$

where

$$f_0(b_2 \mid b_1) \equiv u(q(B)b_2) + \beta \mathbb{E}[u(c_3) \mid B], \quad B = b_1 + b_2.$$

For $b_2 \leq \bar{b}$, the penalty is inactive and the first-order condition is the same as in the no-commitment problem. For an interior breaching choice $b_2 > \bar{b}$, the optimality condition is

$$\frac{\partial}{\partial b_2} f_0(b_2 \mid b_1) = 2\phi(b_2 - \bar{b}), \quad b_2 > \bar{b}. \quad (16)$$

Thus the successor equates the marginal value of additional borrowing to the marginal cost of violating the ceiling. Unlike the fixed-cost case, a quadratic penalty does not generally imply $b_2 = \bar{b}$ in equilibrium: the successor may optimally breach the ceiling by a positive amount when the marginal value of borrowing is sufficiently high.

Let $b_2^Q(b_1, \bar{b})$ denote the policy solving (16). The period-1 government chooses $(b_1, \bar{b})$ anticipating this smooth response. In the region where the ceiling is breached, the optimal ceiling satisfies the envelope condition

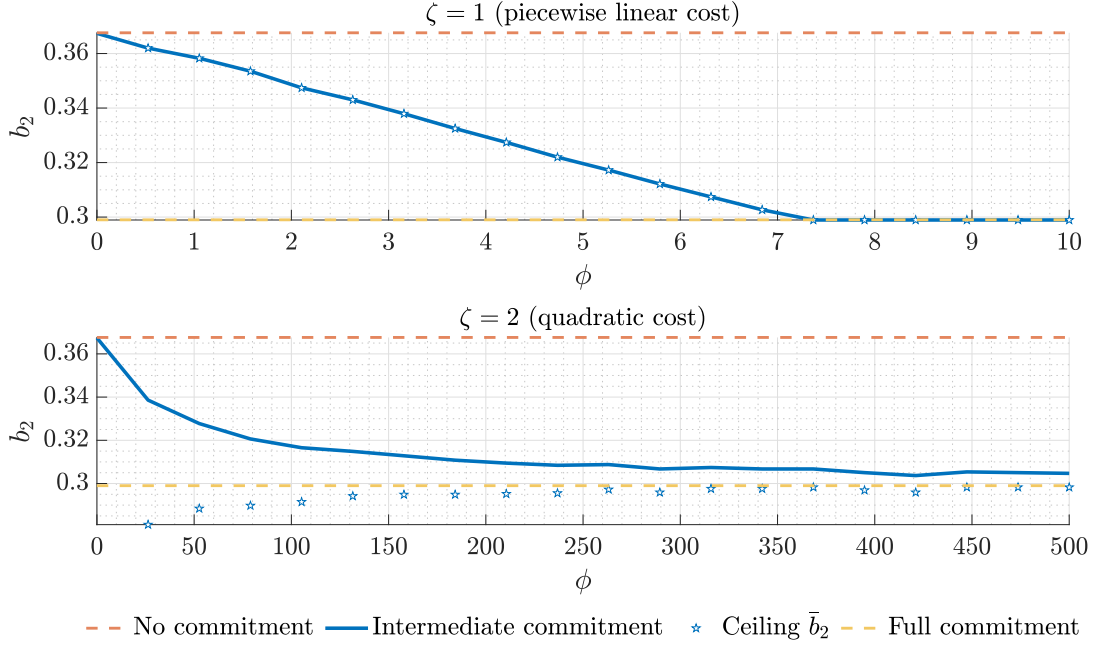
$$u'(c_1) q'(B^Q) b_1 \frac{\partial b_2^Q}{\partial \bar{b}} + 2\beta\phi(b_2^Q - \bar{b}) = 0, \quad (17)$$

where $B^Q = b_1 + b_2^Q(b_1, \bar{b})$ and $c_1 = q(B^Q)b_1$. The first term captures the current pricing cost of relaxing the ceiling: a looser ceiling raises future borrowing, lowers the bond price, and reduces period-1 resources. The second term captures the future flexibility benefit: a looser ceiling reduces the breach cost paid by the successor. The optimal ceiling balances these two forces.

Figure 3 illustrates the implication. Under a quadratic cost, breaches occur smoothly and the gap between borrowing and the ceiling shrinks gradually as ϕ rises. Further derivations for the intermediate-commitment problem with quadratic cost are provided in Appendix A.4.

The behavior of breaches differs across cost specifications. With a quadratic cost, breaches can arise even in the deterministic benchmark because the marginal cost of a small violation is zero at the ceiling. With fixed or linear costs, breaches are absent in the deterministic benchmark: without uncertainty, the value of purchasing flexibility never becomes large enough to justify a discrete violation. In the stochastic extension below, however, adverse states can make the value of extra borrowing high enough to generate breaches even under fixed costs. This distinction motivates the transition to the stochastic environment, where commitment and flexibility interact across states.

Figure 3: Optimal Period-2 Borrowing Under Linear and Quadratic Costs



Note. The figure plots optimal period-2 borrowing b_2 and the associated borrowing ceiling $\bar{b}_2$ as functions of the cost ϕ , under two functional forms: a piecewise-linear penalty ($\zeta = 1$) and a quadratic penalty ($\zeta = 2$). The blue solid line shows optimal constrained borrowing, the blue star markers plot the endogenous ceiling, and the red and yellow dashed lines denote the no-commitment and full-commitment benchmarks, respectively. Increasing enforcement strength progressively disciplines short-term borrowing, with the shape and rate of convergence depending on the curvature of the penalty function.

2.2 A Stochastic Extension: Commitment versus Flexibility

The deterministic benchmark isolates the commitment role of debt ceilings, but it abstracts from state-contingent flexibility. Since period-2 output is known, there is no distinction between states in which the insurance value of relaxing the ceiling is high and states in which it is low. Breaches may still arise under smooth cost specifications, but they do not reflect the rule's ability to provide flexibility selectively in adverse states.

Suppose period-2 output is stochastic,

$$y_2 \in \{y_L, y_H\}, \quad 0 \leq y_L < y_H, \quad \Pr(y_2 = y_L) = \pi,$$

while final output remains deterministic and equal to $\bar{y}$. The period-1 government chooses b_1 and announces a non-contingent ceiling $\bar{b}$ before y_2 is realized. After observing y_s , $s \in \{L, H\}$, the period-2 government chooses $b_2(y_s)$ and pays a deviation cost if $b_2(y_s) > \bar{b}$. Conditional

on state s , define the period-2 objective excluding the ceiling cost as

$$f_0(b_2 \mid b_1, y_s) \equiv u(y_s + q(B)b_2) + \beta \mathbb{E}[u(c_3) \mid B], \quad B = b_1 + b_2,$$

and let

$$b_2^\dagger(b_1, y_s) \in \arg \max_{b_2} f_0(b_2 \mid b_1, y_s)$$

be the unconstrained borrowing choice. Since marginal utility is higher in the low-output state,

$$b_2^\dagger(b_1, y_L) > b_2^\dagger(b_1, y_H).$$

A non-contingent ceiling therefore creates a trade-off: it disciplines borrowing when output is high, but it may be too restrictive when output is low and the consumption-smoothing value of additional borrowing is high.

Consider the fixed-cost case,

$$\Phi(b_2, \bar{b}) = \phi \mathbf{1}_{\{b_2 > \bar{b}\}}.$$

For a given pair $(b_1, \bar{b})$, define the state-specific value of breaching the ceiling by

$$\Delta_s(b_1, \bar{b}) \equiv f_0(b_2^\dagger(b_1, y_s) \mid b_1, y_s) - f_0(\bar{b} \mid b_1, y_s), \quad s \in \{L, H\}.$$

The period-2 government breaches in state s if and only if $\Delta_s(b_1, \bar{b}) > \phi$. Because additional borrowing is more valuable in the low-output state, we have

$$\Delta_L(b_1, \bar{b}) > \Delta_H(b_1, \bar{b}).$$

Proposition 3 gives the central message. When the cost of breaching the ceiling is low, the ceiling is irrelevant; when the cost is high, the ceiling behaves like a rigid borrowing limit. For intermediate costs, the ceiling delivers state-contingent discipline: it constrains borrowing in good times, when flexibility is less valuable, but can be breached in bad times, when the insurance value of additional borrowing is high.

State-Contingent Breaches Under a Fixed Cost

Proposition 3. Fix $(b_1, \bar{b})$ and suppose $\Delta_L(b_1, \bar{b}) > \Delta_H(b_1, \bar{b})$. Under the fixed-cost specification

$$\Phi(b_2, \bar{b}) = \phi \mathbf{1}_{\{b_2 > \bar{b}\}},$$

the period-2 equilibrium has three regions:

No breach: $\phi \geq \Delta_L(b_1, \bar{b}), \quad b_2(y_s) \leq \bar{b} \text{ for } s \in \{L, H\},$

State-contingent breach: $\Delta_H(b_1, \bar{b}) \leq \phi < \Delta_L(b_1, \bar{b}), \quad b_2(y_H) \leq \bar{b}, \quad b_2(y_L) = b_2^\dagger(b_1, y_L),$

Breach in both states: $\phi < \Delta_H(b_1, \bar{b}), \quad b_2(y_s) = b_2^\dagger(b_1, y_s) \text{ for } s \in \{L, H\}.$

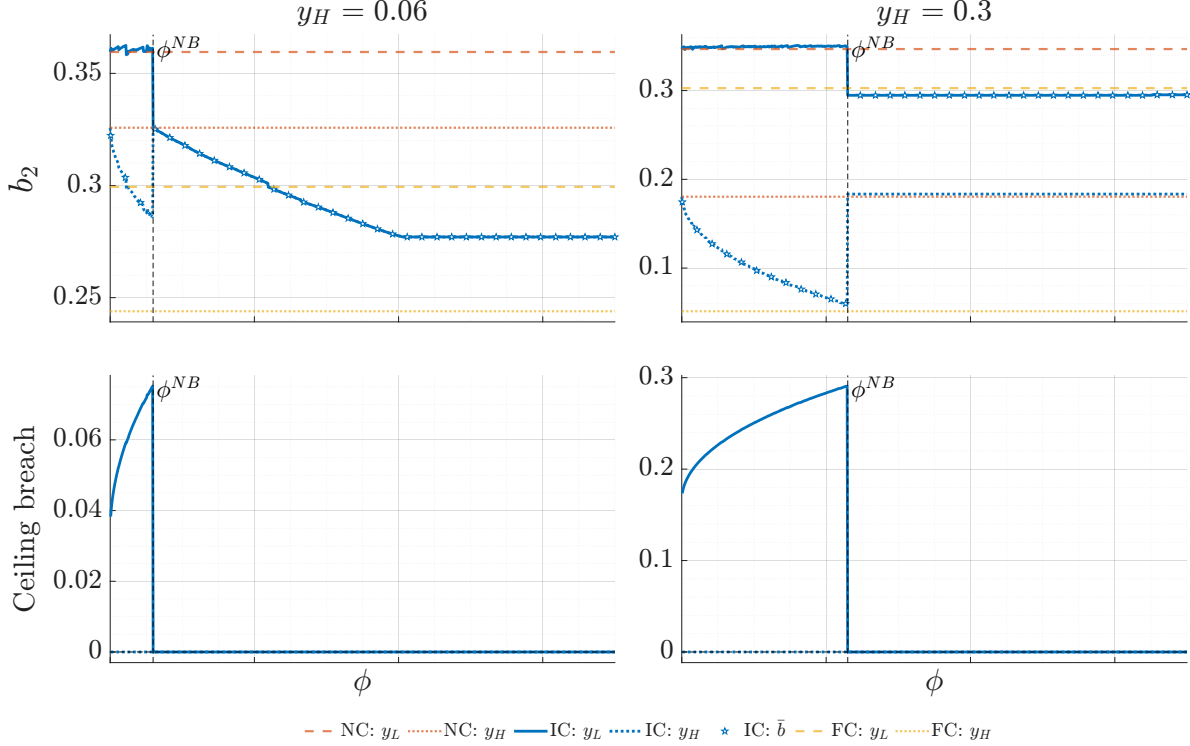
Thus, for intermediate breach costs, the ceiling is respected in the high-output state but breached in the low-output state. Proof. See Appendix A.5.

Proposition 3 characterizes the continuation equilibrium for a given pair $(b_1, \bar{b})$. The ex ante government, however, chooses both initial borrowing and the ceiling, anticipating these state-contingent continuation decisions. As the breach cost varies, the equilibrium therefore moves endogenously across the three regions identified in the proposition.

Figure 4 illustrates this mechanism for two different degrees of output dispersion. In both cases, when breach costs are low, the government uses the ceiling to discipline borrowing only in the high-output state while deliberately preserving flexibility in the low-output state. Consequently, the ceiling is respected following the high realization but breached following the low realization. As the breach cost increases, violating the ceiling in the low-output state becomes progressively more expensive. Beyond the threshold ϕ^{NB} , the government optimally chooses a ceiling that is respected in both states, and the equilibrium enters the no-breach region characterized in Proposition 3.

The comparison across columns shows that greater output dispersion enlarges the state-contingent breach region. Intuitively, larger dispersion increases the difference between the borrowing levels that would be desirable across states. Borrowing is substantially more valuable after the low-output realization, whereas restricting borrowing to obtain some commitment benefits is easier to implement after the high-output realization. Since the government must choose a single non-state-contingent ceiling ex ante, preserving the possibility of breaching the ceiling in the low-output state becomes optimal over a wider range of breach costs.

Figure 4: Debt Ceilings, Borrowing, and State-Contingent Breaches



Note. The figure reports the intermediate-commitment equilibrium for two degrees of period-2 income dispersion, with $y_H = 0.06$ in the left column and $y_H = 0.30$ in the right column, holding $y_L = 0$ and $\Pr(y_L) = \Pr(y_H) = 1/2$. The top row plots state-contingent period-2 borrowing and the endogenous non-state-contingent ceiling as functions of the breach cost ϕ . Solid and dotted blue lines denote borrowing in the low- and high-income states, respectively, and blue markers denote the ceiling; red and yellow lines show the no-commitment and full-commitment benchmarks. The bottom row plots the corresponding ceiling breaches. The vertical dashed line, ϕ^{NB} , denotes the smallest breach cost for which the ceiling is respected in both states.

Finally, the way the government selects the debt ceiling differs with income dispersion. When income dispersion is low (left column), for breach costs below ϕ^{NB} the ceiling binds in the high-income state and is breached in the low-income state. Once breach costs are sufficiently large that breaches disappear, the government selects a ceiling that binds in both states: period-2 borrowing equals the ceiling whether the high- or low-income state is realized. Near the no-breach threshold, the ceiling lies above optimal borrowing under commitment in the low state; for sufficiently large breach costs, it settles between the commitment borrowing levels in the low and high states. Crucially, this ceiling lies below the no-commitment borrowing level in both states and thus confers commitment benefits in both.

When income dispersion is high (right column), by contrast, once breaches disappear the government does so by adopting a less ambitious ceiling. The chosen ceiling lies slightly

below optimal borrowing under commitment in the low-income state.⁹ It binds when the low-income state is realized, so borrowing coincides with the ceiling in that state, but it does not bind in the high-income state. There, the government chooses the same borrowing allocation as under no commitment, since that allocation lies strictly below the ceiling. Thus, to avoid breaches, the government accepts that the ceiling constrains borrowing only in the low-income state.

The government's greater willingness to gamble with tighter ceilings when income risk is low also appears in the infinite-horizon model of Section 3, where we examine its consequences in more detail.

Smooth Breach Costs and the Ceiling Trade-Off. The quadratic case delivers a similar commitment–flexibility trade-off in a smooth marginal form. Suppose

$$\Phi(b_2, \bar{b}) = \phi(b_2 - \bar{b})^2 \mathbf{1}_{\{b_2 > \bar{b}\}}.$$

After observing state s , an interior breaching choice satisfies

$$\frac{\partial}{\partial b_2} f_0(b_2 \mid b_1, y_s) = 2\phi(b_2 - \bar{b}), \quad b_2 > \bar{b}.$$

Thus the government equates the marginal value of additional borrowing to the marginal cost of violating the ceiling, implying larger breaches in low-output states.

The ceiling itself is chosen at time 1, before y_2 is realized. Let $q_1(b_1, \bar{b})$ denote the period-1 debt price induced by the announced ceiling and the associated period-2 borrowing/default incentives, so that $c_1 = q_1(b_1, \bar{b})b_1$. Using the envelope theorem for the period-2 problem, an interior optimal ceiling satisfies

$$\underbrace{-u'(c_1)b_1 q_{1,\bar{b}}(b_1, \bar{b})}_{\text{marginal commitment value of tightening the ceiling}} = \underbrace{\beta \mathbb{E}_s [2\phi(b_2(y_s) - \bar{b}) \mathbf{1}_{\{b_2(y_s) > \bar{b}\}}]}_{\text{marginal expected flexibility cost of tightening the ceiling}}. \quad (18)$$

Equation (18) makes the trade-off transparent. A tighter ceiling raises current debt prices when it reduces expected dilution and default risk. But it also increases expected breach costs in states where the successor wants to borrow above the inherited limit.

An important distinction from the fixed breach cost must be noted, however. With a fixed cost, the benefits of commitment are lost entirely in states where the ceiling is breached:

⁹In numerical experiments, as income dispersion increases, the chosen ceiling approaches the full-commitment borrowing level in the low-income state.

since the penalty does not depend on the size of the breach, lenders expect the government to dilute fully in those states, which weakens the rule. With a quadratic cost, by contrast, the penalty rises with the size of the breach, which limits dilution even in states where the ceiling is breached. This same distinction holds in the model of Section 3, where we examine its consequences in more detail.

3 Infinite-Horizon Model

We build on the canonical [Eaton and Gersovitz \(1981\)](#) framework of sovereign default, incorporating long-term debt and an endogenous debt ceiling. The government chooses both the next period's level of debt and a self-imposed ceiling on future debt, beyond which borrowing incurs a breach cost. This cost is paid only when desired borrowing exceeds the previously announced debt ceiling and captures political, procedural, reputational, or institutional frictions associated with violating the rule. We consider alternative specifications for how this cost varies with the size of the breach.

3.1 Environment

Time is discrete and infinite. In each period, the government observes the realization of an exogenous endowment $y \in \mathcal{Y}$, which follows a Markov process with known transition probabilities. The government begins the period with outstanding debt $B \in \mathbb{R}_+$ and a debt ceiling $\bar{B} \in \mathbb{R}_+$. The debt ceiling is a choice variable in repayment and represents the upper bound above which new borrowing incurs additional costs.

3.2 Government and Lenders

The government chooses whether to repay or default. Let $d \in \{0, 1\}$ be the default indicator, where $d = 1$ denotes default. The government's value function is:

$$V(y, B, \bar{B}) = \max_{d \in \{0, 1\}} (1 - d)V^R(y, B, \bar{B}) + dV^D(y), \quad (19)$$

where V^R is the value of repayment and V^D is the value of default.

Repayment. If the government chooses to repay, it selects next period's debt $B' \in \mathcal{B}$ and a new ceiling $\bar{B}' \in \bar{\mathcal{B}}$. Consumption c satisfies the budget constraint:

$$c + (\delta + (1 - \delta)z)B = y + q(y, B', \bar{B}')[B' - (1 - \delta)B] - \Phi(B', \bar{B}) - \iota(B', B), \quad (20)$$

where $q(y, B', \bar{B}')$ is the price of long-term debt, δ is the fraction of debt maturing each period, z is the coupon, and the function $\iota(B', B)$ captures debt issuance costs. The function $\Phi(B', \bar{B})$ captures the additional cost of borrowing beyond the ceiling:

$$\Phi(B', \bar{B}) = \begin{cases} 0, & \text{if } B' \leq \bar{B}, \\ \phi \cdot (B' - \bar{B})^\zeta, & \text{if } B' > \bar{B}, \end{cases} \quad (21)$$

where $\phi \geq 0$ governs the severity of the penalty. The parameter $\zeta \geq 0$ governs how the political and institutional cost of exceeding the ceiling scales with the size of the breach. Consistent with Section 2, we consider two values for ζ . When $\zeta = 0$, any deviation from the ceiling triggers a single discrete cost ϕ , regardless of how large the breach is. This captures environments where the key friction is a fixed political or procedural hurdle—for example, reopening a fiscal law or bearing a reputational cost that arises once the ceiling is violated, but does not depend on the amount of additional borrowing. In contrast, when $\zeta = 2$, the penalty is convex in the size of the breach, so larger violations become disproportionately more costly. This specification embodies the idea that political resistance, legislative bargaining frictions, or reputational losses intensify with the magnitude of the deviation. Although these two forces may interact in practice, we consider them separately for clarity, allowing us to isolate how discrete versus convex costs affect the credibility and effectiveness of debt-ceiling announcements.

The government's recursive problem in repayment is thus:

$$V^R(y, B, \bar{B}) = \max_{c, B', \bar{B}'} u(c) + \beta \mathbb{E}_{y'} [V(y', B', \bar{B}')], \quad (22)$$

subject to the budget constraint (20).

Default. In the event of default, the country is excluded from financial markets and receives autarky consumption $\varrho(y)$. Re-entry occurs with probability $\theta \in (0, 1)$, in which case the country returns with zero debt and the maximal allowable ceiling $\bar{B}_{\max}$. The default value function is:

$$V^D(y) = u(\varrho(y)) + \beta \mathbb{E}_{y'} [\theta V(y', 0, \bar{B}_{\max}) + (1 - \theta) V^D(y')] . \quad (23)$$

Lenders and bond pricing. There is a continuum of risk-neutral international lenders that discount at the constant gross rate $R = 1 + r$. They are competitive, so in equilibrium the bond price equals the expected discounted payoff per unit of debt. A bond issued at $(y, B', \bar{B}')$ promises, if the government repays next period, a coupon z and the possibility of reselling the remaining $(1 - \delta)$ fraction of the bond. Thus the equilibrium bond price satisfies

$$q(y, B', \bar{B}') = \frac{1}{1 + r} \mathbb{E}_{y'|y} [(1 - d(y', B', \bar{B}')) (\delta + (1 - \delta) (z + q(y', B''(y', B', \bar{B}'), \bar{B}''(y', B', \bar{B}'))))], \quad (24)$$

where $d(y', B', \bar{B}') \in \{0, 1\}$ is the default decision of the next-period government in state $(y', B', \bar{B}')$, and $B''(y', B', \bar{B}')$ and $\bar{B}''(y', B', \bar{B}')$ denote its optimal choices of next-period debt and ceiling. If default occurs ($d = 1$), lenders receive zero, so the payoff in (24) is zero in those states. The price schedule $q(\cdot)$ is therefore determined jointly with the government's policy functions in a recursive Markov-perfect equilibrium.¹⁰

3.3 Calibration

We follow Chatterjee and Eyigungor (2012) in parameterizing preferences, the stochastic endowment process, and the structure of long-term debt. The model is calibrated at a quarterly frequency. The government discounts the future at rate β , households have CRRA utility with coefficient $\sigma = 2$, and international lenders discount at the risk-free rate $r = 0.01$.¹¹ Long-term bonds mature geometrically at rate $\delta = 0.05$ per quarter and pay coupon $z = 0.03$. This implies an average maturity of $1/\delta = 20$ quarters, or five years. Upon default, the economy re-enters financial markets with probability $\theta = 0.0385$ each period.

Endowment process. Output follows an AR(1) process,

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma \varepsilon^2),$$

¹⁰Prices in this environment are fully forward-looking: current lenders care only about future default and borrowing decisions. Under an alternative equilibrium concept, such as that in DAVIS (2019), prices would instead be history-dependent, allowing lenders to punish debt dilution of past commitments. Such punishment would, in turn, reduce the gains from adopting a fiscal rule as a remedy for debt dilution.

¹¹In line with the sovereign default literature, we allow for $\beta < (1 + r)^{-1}$. This wedge captures political myopia, limited commitment, or unmodeled sovereign risks and helps generate realistic debt levels and spreads.

with persistence $\rho = 0.9485$ and innovation standard deviation $\sigma\varepsilon = 0.0271$, identical to the estimates in Chatterjee and Eyigungor (2012).

Default cost function. Following Chatterjee and Eyigungor (2012), the consumption loss in default is given by

$$\varrho(y) = y - \max\{0, d_0 y + d_1 y^2\},$$

where the coefficients (d_0, d_1) determine the magnitude and curvature of the output cost. As discussed below, these parameters are internally calibrated to match key external moments of the Argentine data.

Solution method and issuance costs. Unlike Chatterjee and Eyigungor (2012), we solve the model using the extreme-value shock framework of Dvorkin et al. (2021), which yields smooth policy functions and improves convergence. The correlation and variance of the extreme value shocks are set to $p = 0.37$ and $v = 6.8 \times 10^{-3}$. Because this solution method alters the shape of policy rules relative to the original discrete-choice formulation, we recalibrate the three internally chosen parameters. A detailed description of the computational strategy is reported in Appendix B.

Following Dvorkin et al. (2021), we incorporate debt issuance (adjustment) costs to discipline high-frequency swings in long-term debt. Absent such costs, long-maturity sovereign debt models can generate implausibly large one-period changes in issuance, often followed by immediate default. We specify the issuance cost as

$$\iota(B', B) \equiv \iota_1 \left(\exp(\iota_2 |B' - B|) - 1 \right),$$

and set $\iota_1 = 0.00005$ and $\iota_2 = 28$, as in Dvorkin et al. (2021). This parametrization implies a small marginal cost for routine issuance while making large, discrete adjustments increasingly expensive, helping the model match the smooth debt-accumulation dynamics observed in the data.

Internally calibrated parameters. The parameters β , d_0 , and d_1 are selected to match three targeted moments from the Argentine data: the mean debt-to-output ratio, the mean sovereign spread, and the volatility of spreads. Table 1 reports the targets, model moments, and corresponding parameter values.

Table 1: Targeted Moments and Corresponding Parameters

Moment	Model	Data	Parameter	Value
Debt-to Y ratio	.71	.70	β	0.941
Mean Spread	.0752	.0815	d_0	0.030
Volatility of the Spread	.0408	.0443	d_1	0.431

Note: The table reports the internally calibrated parameters and their corresponding targeted moments.

In what follows, we introduce an endogenous debt ceiling and study its implications under two alternative breach-cost structures: a fixed penalty ($\zeta = 0$) and a convex quadratic penalty ($\zeta = 2$). We first use a relatively high breach cost to illustrate the commitment–flexibility mechanism and show how debt-ceiling announcements affect borrowing incentives, default risk, and sovereign bond prices. We then vary the breach-cost parameter ϕ to quantify how the credibility of the rule shapes long-run spreads, spread volatility, borrowing, breaches, and welfare, and examine how the welfare trade-off changes with aggregate risk. Finally, we isolate the role of state dependence by comparing the endogenous-ceiling economy with a counterfactual in which the ceiling is held constant at its corresponding ergodic mean.

3.4 The Commitment vs Flexibility Mechanism

In the benchmark no-ceiling case with $\phi = 0$ (dashed red line in Figure 5), spreads are determined only by borrowing choices and the exogenous income shock.¹² We then introduce a debt ceiling by setting $\phi = 0.4$, which activates the endogenous penalty $\Phi(B', \bar{B})$ when borrowing exceeds the self-imposed ceiling. The debt ceiling modifies equilibrium outcomes by altering the government’s intertemporal incentives.

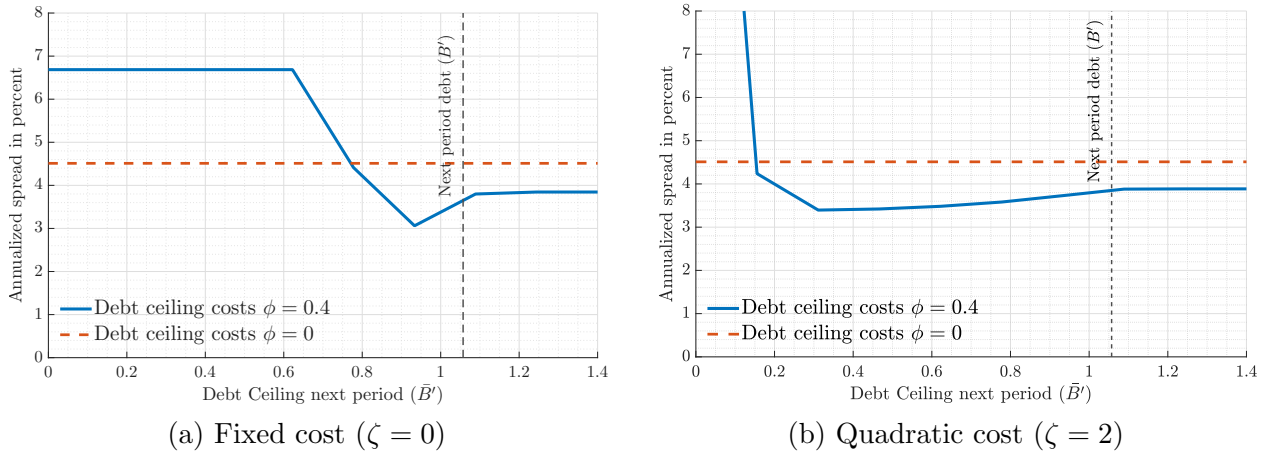
The underlying mechanism hinges on a trade-off between commitment and fiscal flexibility. On the one hand, by choosing and maintaining a ceiling on future borrowing, the government partially commits not to excessively dilute outstanding debt. This mitigates the classic “debt dilution” problem in [Eaton and Gersovitz \(1981\)](#)-type models with long-term debt: when the government issues additional debt today, it lowers the value of legacy bonds tomorrow by raising the probability of default. Anticipating this, investors demand higher yields, which depresses bond prices ([Hatchondo and Martinez, 2009](#); [Chatterjee and Eyigungor, 2012](#)). Promising a credible ceiling can reduce this incentive, raise bond prices, and ultimately lower

¹²If borrowing above the debt ceiling is costless, any debt ceiling announcement is a non-informative signal to future bondholders and as a result the model collapses exactly into the setup of [Chatterjee and Eyigungor \(2012\)](#).

borrowing costs. On the other hand, the same ceiling constrains the government's ability to smooth consumption and respond to adverse shocks by issuing additional debt. In tight fiscal states, this restriction can make default more attractive, increasing the risk of repayment failure.

The net effect of promising a “reasonable” value of the next period's debt ceiling thus reflects the balance of these two forces: lower interest rates in the current period at the cost of more limited borrowing capacity in the next period. Figure 5 illustrates this mechanism.

Figure 5: Debt Ceilings, Sovereign Spreads, and Cost Specifications



Note: Each panel plots the equilibrium annualized spread as a function of the next-period debt ceiling announcement. In both cases, moderate ceilings raise bond prices and lower credit spreads by mitigating debt dilution, whereas excessively tight ceilings increase default risk, thereby reducing prices and widening spreads. Panel **a** assumes a fixed cost of breaching the ceiling ($\zeta = 0$), while Panel **b** assumes a convex quadratic cost ($\zeta = 2$), which makes larger violations disproportionately more costly. Solid blue lines correspond to the model with a high debt-ceiling penalty ($\phi = 0.40$), while the dashed red lines refer to the costless-ceiling case ($\phi = 0$).

The x-axis in Figure 5 reports the level of the promised next-period debt ceiling $\bar{B}'$, while the y-axis shows the annualized credit spreads which are computed as

$$\left(1 + \frac{\delta + (1 - \delta)z}{q(y, B', \bar{B}')} - \delta\right)^4 - (1 + r)^4,$$

corresponding to a fixed level of next-period debt (B' , dashed vertical line). The red dashed line depicts the bond spread in the benchmark model without ceiling frictions ($\phi = 0$). The solid blue line shows the spread in our model where the debt ceiling cost is $\phi = 0.4$. Panel **a** uses a fixed breach cost ($\zeta = 0$) and Panel **b** uses a quadratic breach cost ($\zeta = 2$).

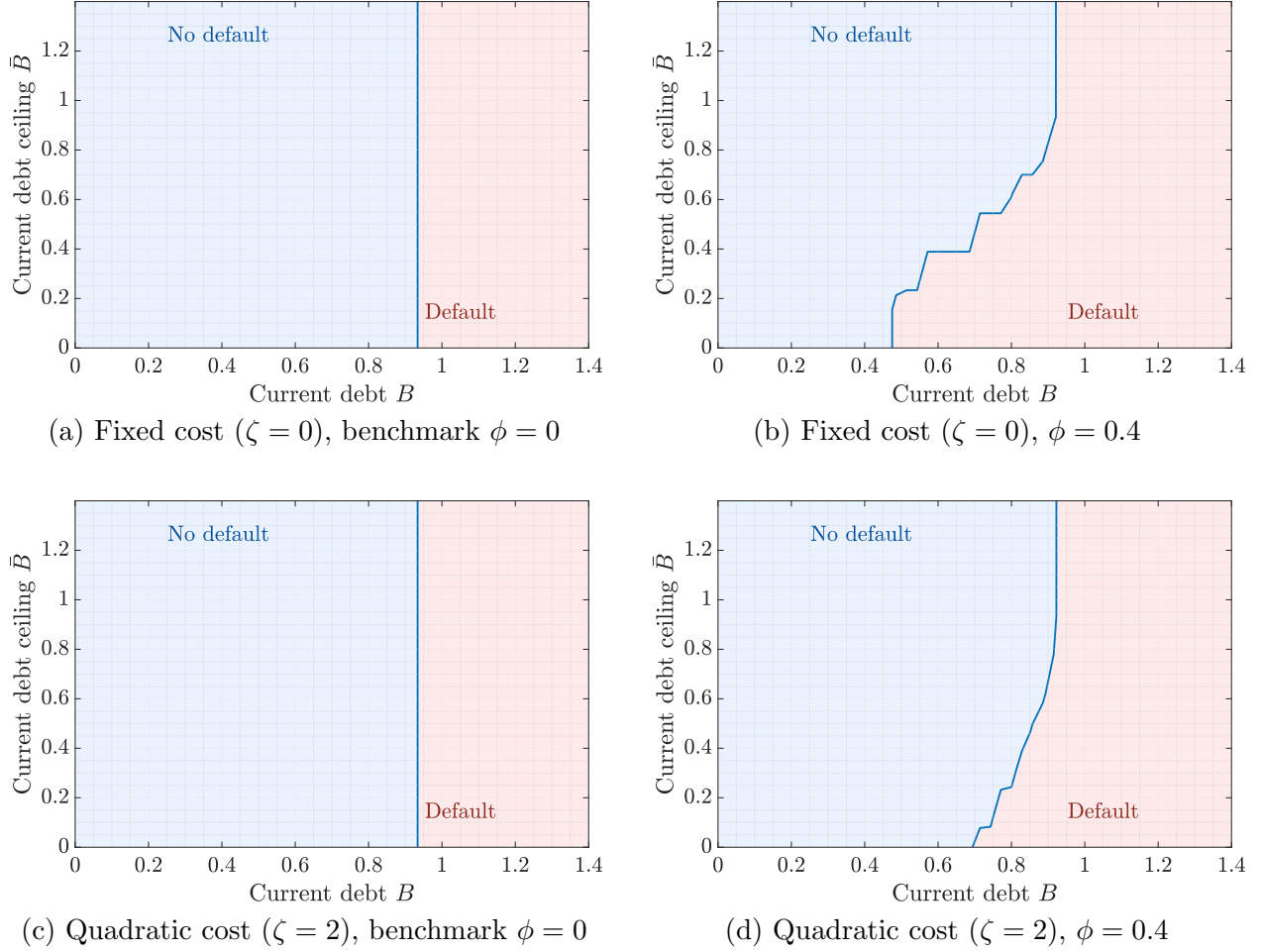
For a wide range of values of $\bar{B}'$ around and moderately below the fixed next-period debt, the announced ceiling raises bond prices and lowers credit spreads relative to the

benchmark without ceiling costs. In this region, the ceiling mitigates debt dilution and lowers borrowing costs. When the ceiling becomes extremely tight, however—well below any level consistent with rolling over existing debt—bond prices fall sharply and credit spreads increase. Very small ceilings imply that the government would almost surely need to violate the ceiling in adverse states, which raises default risk and depresses prices. Under the fixed-cost specification (Panel [a](#)), this shows up as a low, flat segment: once the ceiling is tight enough to make a breach nearly certain, the discrete penalty leads lenders to price in a high probability of default. Under the quadratic specification (Panel [b](#)), the effect is even more pronounced. Because large violations become disproportionately costly, very tight ceilings generate sharply higher default risk and a steep collapse in bond prices. In both cases, excessively low ceilings are counterproductive: instead of reinforcing commitment, they undermine it by making future policy too inflexible to absorb shocks.

In this latter case, the prospect of being unable to respond to future shocks raises the likelihood of default, as shown in Figure [6](#). The associated increase in default risk leads to higher yields and lower bond prices, consistent with Figure [5](#). Figure [6](#) also shows that the default region is larger under fixed costs, since any ceiling breach triggers a discrete penalty and makes repayment unattractive over a broad set of states. By contrast, bond prices fall more sharply under the quadratic specification when ceilings are extremely tight: even if the government continues to repay in many states, lenders price in the possibility of large violations, which become disproportionately costly and raise the ex ante default probability. Hence Figures [5](#) and [6](#) speak to different margins—ex-ante pricing versus the government’s realized default decisions—and together highlight how overly tight ceilings can undermine, rather than reinforce, policy credibility.

Finally, the non-monotonicity reported in Figure [5](#) highlights a central result: promising reasonable austerity can improve credit conditions, while excessive austerity can backfire by undermining repayment incentives. These considerations suggest that the adoption of a debt ceiling can generate welfare gains. By alleviating debt dilution without excessively restricting fiscal flexibility, a moderate debt ceiling can improve the trade-off faced by the government, enabling it to borrow at lower spreads while preserving enough space to respond to adverse shocks. As we show in Subsection [3.6](#), this mechanism can raise ex ante expected utility relative to both the no-ceiling benchmark and overly rigid fiscal rules.

Figure 6: Tight Ceilings Can Trigger Default Under Alternative Cost Structures

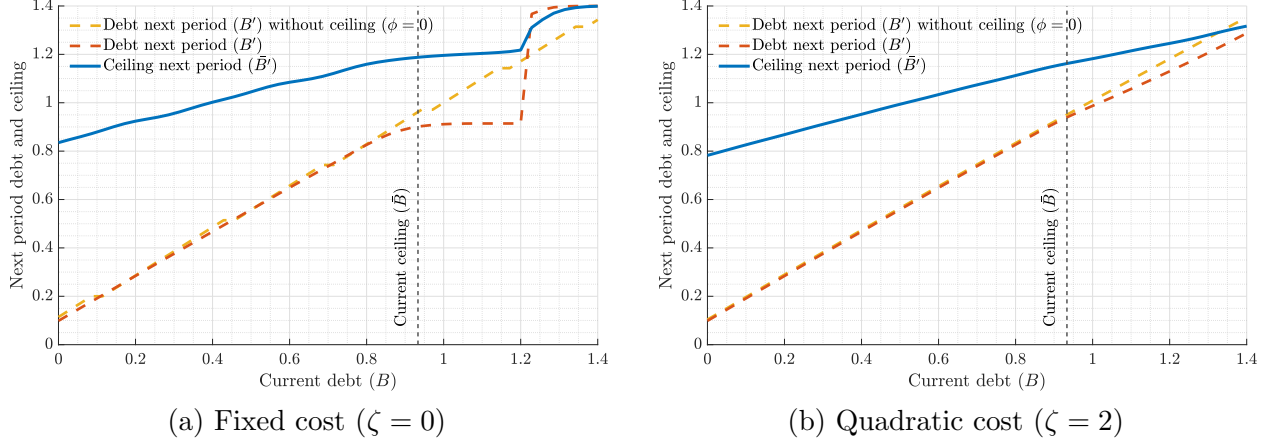


Note: Each panel shows the default region in the space of current debt B and current debt ceiling $\bar{B}$. Tight ceilings can trigger default by weakening repayment incentives. Increasing ϕ from 0 to 0.4 shifts the economy toward higher default risk when the ceiling binds at high debt levels. Fixed costs ($\zeta = 0$) create sharper threshold effects, whereas quadratic costs ($\zeta = 2$) generate smoother distortions in the default boundary.

Finally, we analyze the policy functions for the debt ceiling itself. Figure 7 shows how borrowing and debt-ceiling choices vary with the level of current debt (x-axis), holding the endowment and the inherited ceiling—marked by the vertical dashed line—fixed. In Panel b, which assumes a quadratic breaching cost, the government's adjustment behavior is considerably smoother than under the fixed-cost specification. For current debt levels below the ceiling, borrowing is only mildly disciplined and both next-period debt and the newly announced ceiling move gradually with the state. As current debt approaches and exceeds the ceiling, the marginal cost of violating it rises, keeping next-period debt (the red line) below the unconstrained benchmark (the yellow line) and inducing the government to breach the ceiling (the blue line) cautiously. Unlike the discrete jump observed under the

fixed-cost case in Panel **a**, the quadratic specification produces a continuous response: the ceiling and borrowing choices adjust smoothly, with larger breaches becoming progressively more expensive. Panel **a** indicates that the government finds it optimal to pay the breaching cost only when current debt lies well above the inherited ceiling, at which point it adjusts the announced ceiling upward more decisively.

Figure 7: Debt Ceilings and Borrowing Choices



Note: Each panel plots equilibrium borrowing and debt-ceiling choices as a function of current debt, holding the endowment and the current period ceiling fixed. In both cases, borrowing is partially constrained by the existing ceiling, while future ceilings are chosen slightly above next-period debt. The plots are generated for a high-endowment state in which default risk is nil. Panel **a** assumes a fixed cost of breaching the ceiling ($\zeta = 0$), whereas Panel **b** assumes a convex quadratic cost ($\zeta = 2$). Dashed red and solid blue lines correspond to the model with a high debt-ceiling penalty ($\phi = 0.40$), while the dashed yellow lines refer to the costless-ceiling case ($\phi = 0$).

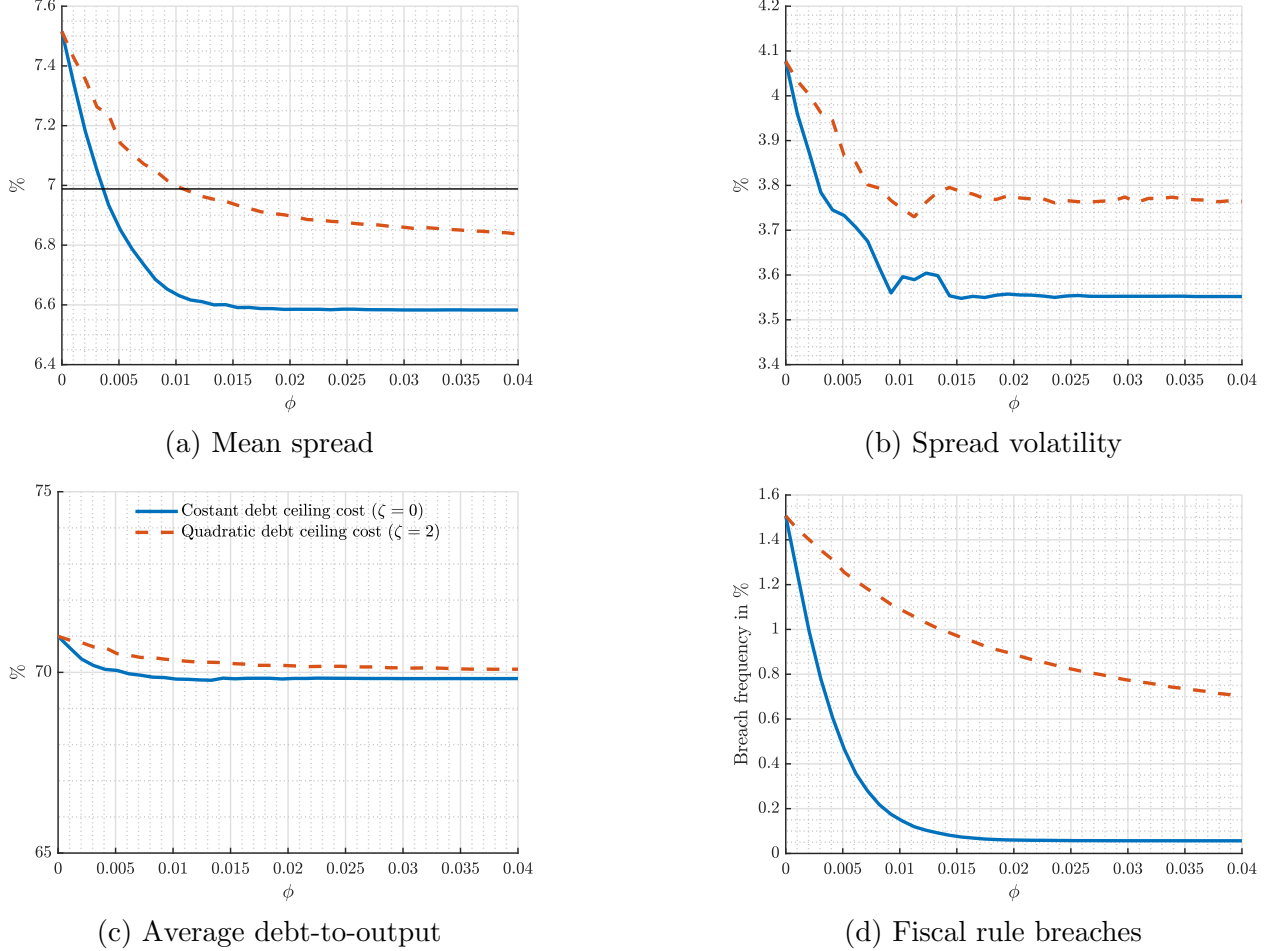
3.5 Debt Ceiling Credibility and Aggregate Outcomes

We next examine how the credibility of the debt ceiling shapes bond pricing and borrowing outcomes. As the cost of breaching the inherited ceiling increases, future issuance is more strongly disciplined, expected dilution falls, and lenders require less compensation for holding sovereign debt. A key result is that these improvements in borrowing terms occur with only a small change in average indebtedness.

Figure 8 reports four ergodic moments as functions of the breach-cost parameter ϕ for both cost specifications. Panel **a** reports the mean annualized spread, Panel **b** spread volatility, Panel **c** the average debt-to-output ratio, and Panel **d** the frequency of ceiling breaches. As ϕ increases, average spreads decline substantially. Under the fixed-cost specification, for example, the mean spread falls from about 7.5 percent in the no-ceiling benchmark to roughly 6.6 percent at the upper end of the grid. Higher breach costs make future deviations from the

announced ceiling less attractive, reducing expected dilution and raising equilibrium bond prices.

Figure 8: Sovereign Outcomes Across Debt-Ceiling Credibility



Note: The figure reports ergodic model moments as functions of the enforcement cost ϕ . Panel **a** plots the average annualized sovereign spread. Panel **b** plots spread volatility, computed as the standard deviation of the quarterly spread and reported in annualized percentage points. Panel **c** plots the average debt-to-output ratio. The horizontal line in Panel **a** marks the no-ceiling model spread minus the estimated reduction in spreads associated with fiscal-rule adoption in the cross-country regressions. Panel **d** plots the annual probability of breaching the debt ceiling. As a reference, at the values of ϕ for which the model spread coincides with the horizontal line in Panel **a**, the average resources spent on enforcing the rule are .08% of output for the fixed-cost model ($\zeta = 0$) and .04% of output for the quadratic-cost model ($\zeta = 2$).

Panel **b** shows that spread volatility also declines as the ceiling becomes more credible. By dampening the response of future borrowing and default risk to the realized state, higher breach costs translate into less volatile sovereign pricing. Under the fixed-cost specification, spread volatility falls from about 4 percent in the no-ceiling benchmark to roughly 3.6 percent at higher values of ϕ .

Importantly, Panel **c** shows that these pricing effects do not rely on a large reduction in average borrowing. The average debt-to-output ratio remains close to 0.70 throughout the grid and declines only modestly as ϕ increases. The ceiling therefore operates primarily by changing the state-contingent incentives governing future issuance rather than by forcing the government to carry substantially less debt on average.

Panel **d** provides a direct measure of the credibility of the announced rule. Breaches become less frequent as ϕ increases, particularly under the fixed-cost specification. Since breaches are costly, the gains from lower spreads must be weighed against the deadweight losses arising from breaches. This trade-off is crucial for the welfare analysis.

Taken together, these results show that greater debt-ceiling credibility can produce sizable improvements in sovereign pricing without materially reducing average debt. These pricing gains, however, need not translate one-for-one into welfare gains, because stronger commitment also reduces fiscal flexibility in adverse states. We turn to this trade-off next.

3.6 Welfare and the Value of Flexibility

We compute the lifetime utility of a patient social planner with discount factor $\beta^{SP} = (1 + r)^{-1}$ under alternative enforcement strengths ϕ , holding the initial state fixed at (y_0, B_0) . The planner evaluates welfare under the *actual* equilibrium behavior of the (impatient) government—that is, taking as given the policy rules $\mathcal{D}_G^\phi$, $\mathcal{B}'_G^\phi$, and $\bar{\mathcal{B}}_G^\phi$ induced by a given ϕ —and discounting forward:

$$V^{SP}(y_0, B_0 \mid \phi) = \sum_{t=0}^{\infty} \frac{u\left[\mathcal{C}\left(\mathcal{D}_G^\phi(t), \mathcal{B}'_G^\phi(t), \bar{\mathcal{B}}_G^\phi(t)\right)\right]}{(1 + r)^t}. \quad (25)$$

We report consumption-equivalent welfare gains relative to the benchmark economy without ceiling frictions ($\phi = 0$). Let $\lambda(\phi)$ denote the constant consumption compensation such that

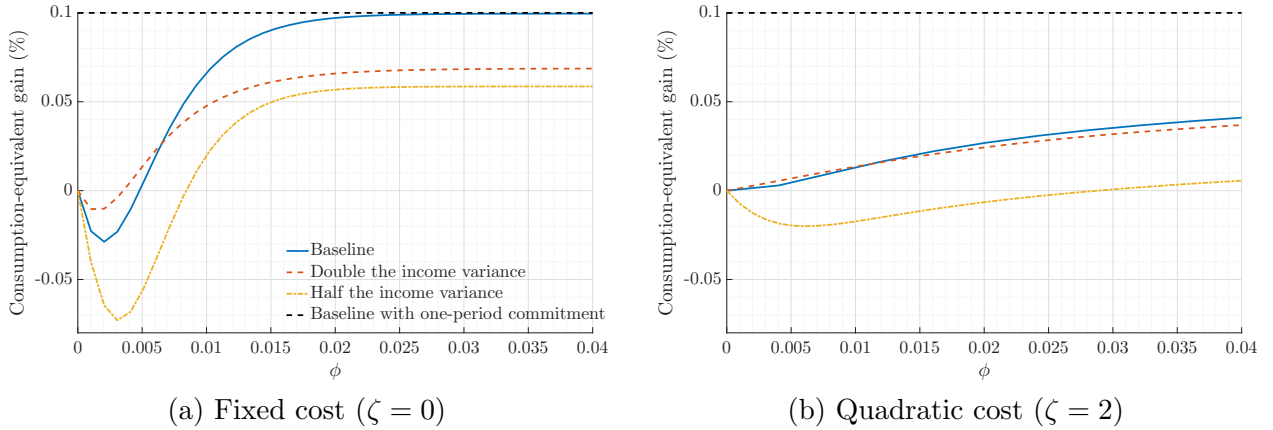
$$V^{SP}(y_0, B_0 \mid \phi) = \sum_{t=0}^{\infty} \frac{u[(1 + \lambda(\phi))c_t^0]}{(1 + r)^t},$$

where $\{c_t^0\}$ is the consumption path implied by the $\phi = 0$ economy starting from the same initial state.

Figure 9 reports consumption-equivalent welfare gains as a function of the breach-cost parameter ϕ and examines how these gains change for different levels of income risk volatility. A central result is that welfare can be non-monotone in ϕ . When deviations remain inexpensive,

the ceiling is frequently breached and provides little commitment, while breach costs are already incurred in equilibrium. In fact, the very high frequency of breaches more than offsets any of the commitment gains the rule provides, leading to overall welfare losses. Because the government discounts the future at $\beta < \beta^{SP}$, it places relatively more weight on the current improvement in borrowing terms generated by a tight ceiling and less weight on the future costs incurred when that ceiling is breached. From the perspective of the more patient planner, the resulting costs of repeated breaches can therefore outweigh the commitment gains. Thus, in welfare terms, intermediate-commitment outcomes need not lie between the no-commitment economy without a rule and the one-period-commitment benchmark: for low breach costs, welfare can be lower than under no commitment. As ϕ increases, the rule becomes more effective at disciplining future issuance, the reduction in expected dilution and borrowing costs dominates these resource costs, and welfare rises. For sufficiently large values of ϕ , breaches disappear and welfare converges asymptotically to the positive one-period-commitment benchmark, shown by the black dashed line.

Figure 9: Welfare Effects of Debt Ceilings and Income Risk



Note: The figure reports consumption-equivalent welfare gains from introducing an endogenous debt ceiling as a function of the breach-cost parameter ϕ . The solid blue lines correspond to the baseline income process, while the dashed red and yellow lines correspond, respectively, to counterfactual economies in which the variance of the income shock is doubled and halved. Welfare gains are measured relative to the corresponding economy with $\phi = 0$. Panel **a** considers a fixed per-breach cost ($\zeta = 0$), while Panel **b** considers a convex quadratic breach cost ($\zeta = 2$).

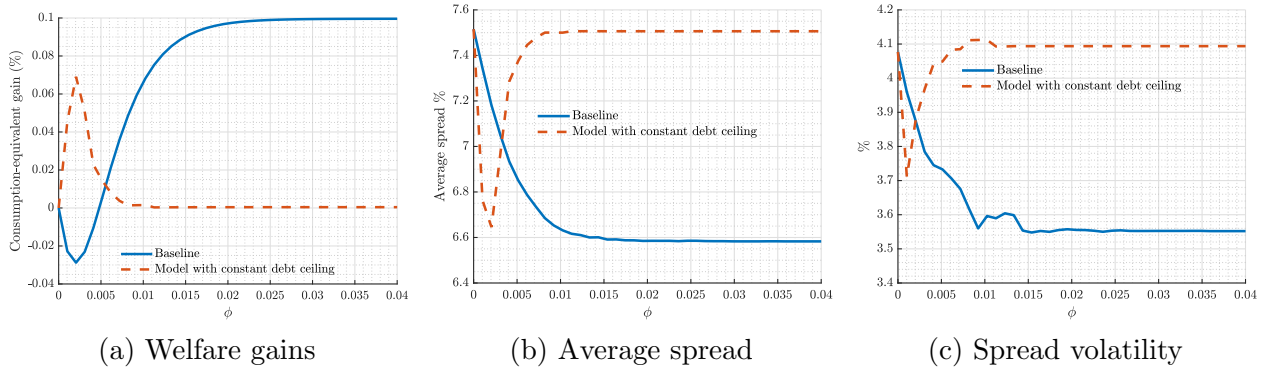
The volatility of income greatly affects the trade-off between commitment gains and fiscal breaches. Doubling the variance of the income shock attenuates the welfare losses at low values of ϕ , but also reduces the gains from large breach costs. When income risk is very volatile, the government is more reluctant to propose rules that it knows are likely to be breached. As a result, the set of values of ϕ for which the costs of breaches surpass the gains from commitment is smaller. Conversely, when income risk is low, the government's willingness to

gamble with a rule that will be breached in certain states of the world increases, and thus the region with welfare losses expands. In the fixed-cost economy, the initial welfare trough becomes substantially shallower, while the gains at high ϕ converge to a noticeably lower level. The same compression of the welfare profile is visible under quadratic costs: greater income risk largely eliminates the small local non-monotonicities at low ϕ , but eventually produces lower welfare gains than in the baseline economy as the breach cost increases.¹³ Thus, greater aggregate risk makes welfare less sensitive to the strength of the breach penalty and reduces the payoff from moving toward very strong commitment.

3.7 Why Endogenous Ceilings Matter

Figure 10 isolates the value of allowing the debt ceiling to adjust endogenously to the state. For each value of the breach-cost parameter ϕ , we first solve the baseline model with an endogenous ceiling and compute its ergodic mean. We then solve a counterfactual economy in which the ceiling is fixed permanently at that same ϕ -specific mean, while the government retains the option to exceed it by paying the same breach cost. The comparison therefore holds fixed the average level of the ceiling and removes only its state dependence.

Figure 10: Endogenous versus Constant Debt Ceilings



Note: The figure compares the baseline model with an endogenous debt ceiling to a counterfactual economy in which the ceiling is exogenous and constant over time, while the government may still exceed it by paying the breach cost. For each value of ϕ , we first solve the endogenous-ceiling model with a fixed breach cost ($\zeta = 0$) and compute the ergodic mean of the debt ceiling. We then solve the corresponding counterfactual economy holding the debt ceiling fixed at this ϕ -specific ergodic mean. This procedure is repeated separately for each value of ϕ in the grid. Panel **a** reports consumption-equivalent welfare gains, Panel **b** reports the average sovereign spread, and Panel **c** reports spread volatility.

¹³Welfare losses under a quadratic breach cost tend to be smaller than under a fixed cost because, as discussed in Section 2.2, breaches involve smaller debt deviations and therefore less dilution. At the same time, small breaches remain relatively inexpensive under a quadratic cost even for large values of ϕ , so welfare losses can arise over a wider range of parameter values.

The difference between the two economies is quantitatively relevant. Panel [a](#) shows that the constant-ceiling economy delivers positive welfare gains for a range of low values of ϕ . In the baseline model, over the same range of low breach costs, the government repeatedly chooses tight ceilings that it subsequently breaches, with breach frequencies particularly high in this region (Panel [d](#) of Figure 8), leading to welfare losses. Fixing the ceiling prevents this behavior and thereby avoids the costs associated with frequent violations. These gains disappear as breaching the rule becomes more costly. For high breach costs, the fixed ceiling becomes too loose to provide meaningful commitment, and the constant-ceiling economy approaches the no-commitment benchmark. By contrast, welfare in the endogenous-ceiling economy rises substantially and then levels off. Average spreads and spread volatility display a similar pattern. Panel [b](#) shows that the baseline model with an endogenous ceiling produces a persistent decline in average sovereign spreads, whereas under a constant ceiling spreads eventually return close to the no-ceiling level. Panel [c](#) shows the analogous pattern for spread volatility. The role of state dependence therefore depends on the credibility of the rule: at low breach costs, limiting adjustment can improve welfare by preventing tight ceilings that are subsequently breached, whereas at high breach costs endogenous adjustment is essential for generating meaningful commitment gains.

These results can be understood through the commitment–flexibility trade-off documented in Figure 5. Holding next-period debt fixed, Figure 5 shows that spreads are non-monotone in the announced ceiling. Moderately restrictive ceilings lower spreads because they limit future dilution. Ceilings that are too tight instead raise spreads because they restrict the government’s ability to respond to adverse shocks and increase future default risk. The corresponding expansion of the default region is visible in Figure 6. Thus, there is no single ceiling level that provides the same balance between commitment and fiscal flexibility in every state.

The endogenous-ceiling economy allows the government to adjust the ceiling to the state of the economy. Figure 7 shows that the chosen next-period ceiling moves systematically with current debt and with planned next-period borrowing. In particular, the government generally chooses $\bar{B}'$ above its planned debt position B' , preserving some borrowing capacity while still placing a meaningful restriction on subsequent issuance. The full policy functions reported in Appendix Figure 11 reinforce this point: the optimal ceiling varies substantially with current debt, the inherited ceiling, and the income realization. The fiscal rule is therefore not characterized by a single preferred debt limit; its appropriate level changes with the fiscal state.

This state dependence also clarifies why the relative performance of the two economies

depends on the breach cost. When breach costs are low, allowing the ceiling to adjust state by state can induce the government to choose tight ceilings that it subsequently breaches, so restricting this adjustment can improve welfare. When breach costs are high, instead, endogenous adjustment allows the government to choose ceilings that are sufficiently tight to provide meaningful commitment, whereas a ceiling fixed at its ergodic mean can be too loose. More generally, a constant ceiling can be too loose in states in which limiting future issuance is especially valuable and too tight in states in which fiscal flexibility is valuable.

The two economies can be interpreted as limiting cases of a model in which changing the inherited ceiling is itself costly. For example, suppose adjustments in the ceiling were subject to a cost $\frac{\kappa}{2}(\bar{B}' - \bar{B})^2$. Our baseline corresponds to $\kappa = 0$, so that the announced ceiling can adjust freely with the state, whereas the constant-ceiling counterfactual corresponds to the limiting case $\kappa \rightarrow \infty$, after initializing the ceiling at the corresponding ergodic mean. Intermediate values of κ would generate partially adjustable fiscal rules between these two benchmarks. In this sense, the counterfactual isolates the role of state dependence in the design of the debt limit.

Debt ceilings in crises. The state dependence documented above becomes particularly relevant in adverse states. Appendix Figure 12 reports an event analysis around fiscal crises, defined in the no-ceiling economy as episodes in which the sovereign spread rises more than two standard deviations above its mean. We then expose the economy with an endogenous debt ceiling and a fixed breach cost of $\phi = 0.04$ to the same sequence of output shocks. The exercise therefore compares the equilibrium responses of the two economies to a common deterioration in fundamentals.

The endogenous ceiling rises sharply at the onset of the crisis, providing additional fiscal space precisely when borrowing needs increase. This response is the dynamic counterpart of the commitment–flexibility trade-off in Figure 5: rather than maintaining a tight limit when doing so would increase repayment risk, the government relaxes the announced ceiling in the adverse state. Importantly, this flexibility does not eliminate the discipline provided by the rule. Relative to the no-ceiling economy, the spike in sovereign spreads is substantially attenuated, while the increase in debt and the decline in consumption are only modestly smaller. Thus, the quantitative gains do not come primarily from a large reduction in debt accumulation: the debt paths remain relatively close, differing by only about 2 percentage points of GDP at the peak of the crisis, while borrowing costs differ substantially. Crises are precisely the periods in which borrowing needs are high and debt sustainability depends critically on borrowing costs. The debt ceiling lowers spreads by increasing the credibility of

the fiscal consolidation that follows the crisis.

4 Conclusion

This paper develops a positive theory of self-imposed debt ceilings as intermediate commitment devices in sovereign debt markets. A debt ceiling disciplines future borrowing and limits debt dilution, but can be breached at a cost when fiscal flexibility becomes sufficiently valuable. A three-period model isolates this trade-off. Under discretion, borrowing is tilted toward the future because successor governments do not internalize the dilution of legacy bondholders. A costly-to-breach ceiling partially restores commitment, while uncertainty makes the effective degree of commitment state-dependent: the same ceiling can discipline borrowing in favorable states and be breached in adverse ones. For moderate or sufficiently high breach costs, greater income volatility limits how ambitious a ceiling the government can credibly set.

In the infinite-horizon model, more credible ceilings substantially lower sovereign spreads and spread volatility while changing average debt only modestly. The welfare effects, however, are non-monotone. When breach costs are low, governments choose tight ceilings that they subsequently violate frequently, and the resulting costs can make welfare lower than in the no-commitment economy without a rule. Intermediate commitment therefore need not lie, in welfare terms, between no commitment and the one-period-commitment benchmark. As breach costs rise, violations become less frequent, dilution falls, and welfare converges toward the positive one-period-commitment gain. The ability to adjust the ceiling with economic conditions is also quantitatively important: at low breach costs, restricting this flexibility through a constant ceiling can improve welfare by preventing repeated breaches, whereas at high breach costs endogenous adjustment is important for realizing commitment gains. During sovereign crises, ceilings provide fiscal space while substantially attenuating the increase in borrowing costs, in part by increasing the credibility of the subsequent fiscal consolidation.

These results suggest that the appropriate degree of flexibility in a fiscal rule depends on its credibility. When rules are easily breached, additional flexibility can encourage promises that are repeatedly abandoned, and a simpler framework with less scope for adjustment may perform better. When rules are sufficiently credible, state-contingent adjustment instead allows governments to combine meaningful commitment with flexibility in adverse states. Fiscal-rule design therefore involves not only how tight a limit is, but also how costly it is to violate and how easily it can be adjusted as economic conditions change.

References

- Aguiar, Mark, Manuel Amador, and Stelios Fourakis**, “On the welfare losses from external sovereign borrowing,” *IMF Economic Review*, 2020, *68* (1), 163–194.
- Alesina, A., C. Favero, and F. Giavazzi**, “The Output Effect of Fiscal Consolidations,” *Journal of International Economics*, 2015, *96* (S1), 19–42.
- Amador, Manuel, Iván Werning, and George-Marios Angeletos**, “Commitment vs. flexibility,” *Econometrica*, 2006, *74* (2), 365–396.
- Arce, Fernando and Stelios Fourakis**, “Fiscal consolidations and Political Frictions,” *Working paper*, 2025.
- Asonuma, Tamon and Christoph Trebesch**, “Sovereign debt restructurings: preemptive or post-default,” *Journal of the European Economic Association*, 2016, *14* (1), 175–214.
- , **Dirk Niepelt, and Romain Ranciere**, “Sovereign bond prices, haircuts and maturity,” *Journal of international economics*, 2023, *140*, 103689.
- Athey, Susan, Andrew Atkeson, and Patrick J. Kehoe**, “The Optimal Degree of Discretion in Monetary Policy,” *Econometrica*, 2005, *73* (5), 1431–1475.
- Bocola, Luigi, Gaston Chaumont, Alessandro Dovis, and Rishabh Kirpalani**, “Accounting for Credibility: Fiscal-Monetary Interactions and the Credibility of Central Bank Mandates,” February 2026. Working paper.
- Chatterjee, Satyajit and Burcu Eyigungor**, “Maturity, indebtedness, and default risk,” *American Economic Review*, 2012, *102* (6), 2674–2699.
- Clymo, Alex and Andrea Lanteri**, “Fiscal Policy with Limited-Time Commitment,” *Economic Journal*, 2020, *130* (627), 623–652.
- , —, and **Alessandro T. Villa**, “Capital and labor taxes with costly state contingency,” *Review of Economic Dynamics*, 2023, *51*, 943–964.
- Cruces, Juan J. and Christoph Trebesch**, “Sovereign Defaults: The Price of Haircuts,” *American Economic Journal: Macroeconomics*, July 2013, *5* (3), 85–117.
- Debortoli, D. and R. Nunes**, “Fiscal policy under loose commitment,” *Journal of Economic Theory*, 2010, *145* (3), 1005–1032.

- **and** — , “Lack of Commitment and the Level of Debt,” *Journal of the European Economic Association*, 2013, *11* (5), 1053–1078.
- Dovis, Alessandro**, “Efficient sovereign default,” *The Review of Economic Studies*, 2019, *86* (1), 282–312.
- Dvorkin, Maximiliano, Juan M Sánchez, Horacio Sapriza, and Emircan Yurdagul**, “Sovereign debt restructurings,” *American Economic Journal: Macroeconomics*, 2021, *13* (2), 26–77.
- Eaton, Jonathan and Mark Gersovitz**, “Debt with potential repudiation: Theoretical and empirical analysis,” *The Review of Economic Studies*, 1981, *48* (2), 289–309.
- Espino, Emilio, Julian Kozlowski, Fernando M Martin, and Juan M Sánchez**, “Policy rules and large crises in emerging markets,” *FEB St. Louis Working Paper*, 2022, (2022-18).
- Farhi, Emmanuel**, “Capital Taxation and Ownership When Markets are Incomplete,” *Journal of Political Economy*, 2010, *118* (5), 908–948.
- Fernández-Villaverde, Jesús, Pablo Guerrón-Quintana, Keith Kuester, and Juan Rubio-Ramírez**, “Fiscal Volatility Shocks and Economic Activity,” *American Economic Review*, 2015, *105* (11), 3352–3384.
- Halac, Marina and Pierre Yared**, “Fiscal rules and discretion under persistent shocks,” *Econometrica*, 2014, *82* (5), 1557–1614.
- **and** — , “Fiscal rules and discretion under self-enforcement,” Technical Report, National Bureau of Economic Research 2017.
- **and** — , “Commitment versus flexibility with costly verification,” *Journal of Political Economy*, 2020, *128* (12), 4523–4573.
- **and** — , “Fiscal rules and discretion under limited enforcement,” *Econometrica*, 2022, *90* (5), 2093–2127.
- Hatchondo, Juan Carlos and Leonardo Martinez**, “Long-duration bonds and sovereign defaults,” *Journal of international Economics*, 2009, *79* (1), 117–125.
- , — , **and César Sosa-Padilla**, “Debt dilution and sovereign default risk,” *Journal of Political Economy*, 2016, *124* (5), 1383–1422.

- , – , and **Francisco Roch**, “Fiscal rules and the sovereign default premium,” *American Economic Journal: Macroeconomics*, 2022, *14* (4), 244–273.
- , – , and – , “Numerical fiscal rules for economic unions: The role of sovereign spreads,” *Economics Letters*, 2022, *210*, 110168.
- Karantounias, Anastasios G.**, “Greed versus Fear: Optimal Time-Consistent Taxation with Default,” Technical Report 2019.
- King, Robert G., Charles I. Plosser, and Sergio T. Rebelo**, “Production, Growth, and Business Cycles, II. New Directions,” *Journal of Monetary Economics*, 1988, *21*, 309–341.
- Klein, Paul, P. Krusell, and José V. Ríos-Rull**, “Time-Consistent Public Policy,” *Review of Economic Studies*, 2008, *75* (3), 789–808.
- Mateos-Planas, Xavier, Sean McCrory, José-Víctor Ríos-Rull, and Adrien Wicht**, “Commitment in the canonical sovereign default model,” *Journal of International Economics*, 2025, *157*, 104120.
- Mertens, K. and M. Ravn**, “Understanding the Effects of Anticipated and Unanticipated Tax Policy Shocks,” *Review of Economic Dynamics*, 2011, *14* (1), 27–54.
- and – , “Empirical evidence on the aggregate effects of anticipated and unanticipated U.S. tax policy shocks,” *American Economic Journal: Economic Policy*, 2012, *4* (2), 145–181.
- Roch, Francisco and Francisco Roldán**, “Uncertainty premia, sovereign default risk, and state-contingent debt,” *Journal of Political Economy Macroeconomics*, 2023, *1* (2), 334–370.
- Schmitt-Grohe, S. and M. Uribe**, “Balanced-Budget Rules, Distortionary Taxes, and Aggregate Instability,” *Journal of Political Economy*, 1997, *105* (5), 976–1000.

A Mathematical Details of the Three-Period Model

This appendix provides derivations for the three-period benchmark in Section 2. We first derive expected utility at $t = 3$ and the pricing function, then characterize the no-commitment and commitment allocations, and finally analyze the intermediate-commitment case with a debt ceiling.

A.1 Environment and Default

Time is $t \in \{1, 2, 3\}$. The government receives endowments $y_1 = y_2 = 0$ and $y_3 = \bar{y} > 0$. Preferences are CRRA with risk aversion $\sigma = 2$:

$$u(c) = -\frac{1}{c}, \quad \beta \in (0, 1].$$

All debt matures at $t = 3$. Let b_1 denote face value issued at $t = 1$ and b_2 face value issued at $t = 2$, so total obligations at $t = 3$ are

$$B = b_1 + b_2.$$

Following the main text, define the repayment slack

$$x \equiv \frac{\bar{y} - B}{\bar{y}} \in (0, 1], \quad \text{so that} \quad B = \bar{y}(1 - x).$$

Default cost at $t = 3$ is governed by a Pareto random variable $\theta \sim \text{Pareto}(\alpha)$ on $[1, \infty)$ with shape parameter $\alpha = \frac{1}{2}$. If the government repays, it consumes $c_3 = \bar{y} - B = \bar{y}x$. If it defaults, it consumes $c_3 = \bar{y}/\theta$. Default occurs whenever

$$\frac{\bar{y}}{\theta} \geq \bar{y} - B \iff \theta \leq \frac{\bar{y}}{\bar{y} - B} = \frac{1}{x}.$$

With risk-neutral lenders and a unit gross risk-free rate, the bond price equals the probability of repayment. Using the CDF of the Pareto distribution,

$$q(B) = \Pr(\theta > 1/x) = x^\alpha = \sqrt{x}.$$

Expected utility at $t = 3$ conditional on B is

$$\begin{aligned}
\mathbb{E}[u(c_3) \mid B] &= \Pr(\theta > 1/x) u(\bar{y} - B) + \int_1^{1/x} u\left(\frac{\bar{y}}{\theta}\right) \alpha \theta^{-(\alpha+1)} d\theta \\
&= x^{1/2} \left(-\frac{1}{\bar{y}x} \right) - \int_1^{1/x} \frac{1}{\bar{y}} \frac{1}{2} \theta^{-1/2} d\theta \\
&= -\frac{x^{1/2}}{x\bar{y}} - \frac{1}{\bar{y}} \left[\theta^{1/2} \right]_1^{1/x} \\
&= -\frac{x^{1/2}}{x\bar{y}} - \frac{1}{\bar{y}} \left(x^{-1/2} - 1 \right) \\
&= \frac{1}{\bar{y}} \left(1 - 2x^{-1/2} \right),
\end{aligned}$$

which is equation (4) in the main text.

A.2 No Commitment

We first solve for the period-2 best response $b_2^\dagger(b_1)$ taking b_1 as given, and then characterize the period-1 choice b_1^{NC} .

Period-2 Problem

Given b_1 , total debt is $B = b_1 + b_2$, the slack is $x = 1 - (b_1 + b_2)/\bar{y}$, and the bond price is $q(B) = \sqrt{x}$. Period-2 consumption is

$$c_2 = q(B) b_2 = \sqrt{x} b_2.$$

The period-2 government chooses b_2 to maximize

$$\begin{aligned}
V_2^{NC}(b_1) &= \max_{b_2} \left\{ u(c_2) + \beta \mathbb{E}[u(c_3) \mid B] \right\} \\
&= \max_{b_2} \left\{ -\frac{1}{\sqrt{x} b_2} + \frac{\beta}{\bar{y}} \left(1 - 2x^{-1/2} \right) \right\},
\end{aligned}$$

where $x = 1 - (b_1 + b_2)/\bar{y}$ and $dx/db_2 = -1/\bar{y}$.

The first-order condition (FOC) is

$$\frac{\partial V_2^{NC}}{\partial b_2} = \frac{x^{-1/2}}{b_2^2} - \frac{1}{2\bar{y}} \frac{x^{-3/2}}{b_2} - \frac{\beta}{\bar{y}^2} x^{-3/2} = 0.$$

Multiplying by $x^{3/2}\bar{y}^2b_2^2$ and using $x = 1 - (b_1 + b_2)/\bar{y}$, this simplifies to

$$-\frac{b_2}{2\bar{y}} + x - \frac{\beta b_2^2}{\bar{y}^2} = 0.$$

Substituting $x = 1 - (b_1 + b_2)/\bar{y}$ and rearranging yields the quadratic in $b_2/\bar{y}$:

$$\beta\left(\frac{b_2}{\bar{y}}\right)^2 + \frac{3}{2}\left(\frac{b_2}{\bar{y}}\right) + \left(\frac{b_1}{\bar{y}} - 1\right) = 0. \quad (26)$$

We take the positive root and define

$$b_2^\dagger(b_1) = \frac{\bar{y}}{4\beta} \left[-3 + \sqrt{9 + 16\beta\left(1 - \frac{b_1}{\bar{y}}\right)} \right], \quad (27)$$

which coincides with equation (10) in the main text.

Using (26), the associated slack can be written as

$$x^\dagger(b_1) = 1 - \frac{b_1 + b_2^\dagger(b_1)}{\bar{y}} = \beta\left(\frac{b_2^\dagger(b_1)}{\bar{y}}\right)^2 + \frac{1}{2}\left(\frac{b_2^\dagger(b_1)}{\bar{y}}\right). \quad (28)$$

Differentiating (27) with respect to b_1 gives

$$\frac{db_2^\dagger}{db_1}(b_1) = -\frac{2}{\sqrt{9 + 16\beta\left(1 - \frac{b_1}{\bar{y}}\right)}} = -\frac{2}{4\beta\frac{b_2^\dagger(b_1)}{\bar{y}} + 3}. \quad (29)$$

From (28) we also obtain

$$\frac{dx^\dagger}{db_1}(b_1) = -\frac{1 + \frac{db_2^\dagger}{db_1}(b_1)}{\bar{y}}. \quad (30)$$

The value of following the optimal no-commitment policy at $t = 2$ is

$$V_2^{NC}(b_1) = -\frac{(x^\dagger(b_1))^{-1/2}}{b_2^\dagger(b_1)} + \frac{\beta}{\bar{y}}\left(1 - 2(x^\dagger(b_1))^{-1/2}\right). \quad (31)$$

Differentiating (31) and using (29)–(30) yields

$$\frac{dV_2^{NC}}{db_1}(b_1) = -\left(1 + \frac{db_2^\dagger}{db_1}\right)\frac{(x^\dagger)^{-3/2}}{\bar{y}}\left[\frac{1}{2b_2^\dagger} + \frac{\beta}{\bar{y}}\right] + (x^\dagger)^{-1/2}\frac{\frac{db_2^\dagger}{db_1}}{(b_2^\dagger)^2}, \quad (32)$$

where $x^\dagger = x^\dagger(b_1)$ and $b_2^\dagger = b_2^\dagger(b_1)$.

Period-1 Problem

Anticipating the period-2 policy $b_2^\dagger(b_1)$, the period-1 government solves

$$V_1^{NC}(b_1) = u(c_1) + \beta V_2^{NC}(b_1) = -\frac{1}{\sqrt{x^\dagger(b_1)} b_1} + \beta V_2^{NC}(b_1),$$

where $x^\dagger(b_1)$ and $V_2^{NC}(b_1)$ are given in (28) and (31). The FOC is

$$\begin{aligned} 0 &= \frac{dV_1^{NC}}{db_1}(b_1) \\ &= -\left[\frac{d}{db_1}((x^\dagger)^{-1/2}) \cdot \frac{1}{b_1} + x^{\dagger-1/2} \cdot \frac{d}{db_1}\left(\frac{1}{b_1}\right) \right] + \beta \frac{dV_2^{NC}}{db_1}(b_1). \end{aligned}$$

Using

$$\frac{d}{db_1}((x^\dagger)^{-1/2}) = \frac{1 + \frac{db_2^\dagger}{db_1}}{2\bar{y}} (x^\dagger)^{-3/2}, \quad \frac{d}{db_1}\left(\frac{1}{b_1}\right) = -\frac{1}{b_1^2},$$

and substituting (32), after simplification we obtain

$$0 = (x^\dagger) \left[1 - \beta \frac{\left(\frac{b_1}{\bar{y}}\right)^2}{\left(\frac{b_2^\dagger(b_1)}{\bar{y}}\right)^2} \right] - \frac{1}{2} \left(\frac{b_1}{\bar{y}}\right) \frac{4\beta \left(\frac{b_2^\dagger(b_1)}{\bar{y}}\right) + 1}{4\beta \left(\frac{b_2^\dagger(b_1)}{\bar{y}}\right) + 3}. \quad (33)$$

(33) together with (27) characterizes the no-commitment allocation $(b_1^{NC}, b_2^{NC}) = (b_1^*, b_2^*)$ in the main text.

Proof of Proposition 1

We now prove the maturity comparison in Proposition 1, reproduced here for convenience:

Fix $\beta \in (0, 1)$ and $\bar{y} > 0$. Let (b_1^C, b_2^C) be the commitment allocation and (b_1^{NC}, b_2^{NC}) the no-commitment allocation. Then

$$\frac{b_2^{NC}}{b_1^{NC} + b_2^{NC}} \geq \frac{b_2^C}{b_1^C + b_2^C} = \frac{\sqrt{\beta}}{1 + \sqrt{\beta}}.$$

Under commitment, the FOCs (derived below) imply $b_2^C/b_1^C = \sqrt{\beta}$. Suppose, for a contradiction, that the no-commitment allocation were relatively more front-loaded:

$$\frac{b_2^C}{b_1^C} = \sqrt{\beta} > \frac{b_2^{NC}}{b_1^{NC}}.$$

Equivalently,

$$\beta > \left(\frac{b_2^{NC}}{b_1^{NC}} \right)^2 = \left(\frac{\frac{b_2^{NC}}{\bar{y}}}{\frac{b_1^{NC}}{\bar{y}}} \right)^2 \implies 1 - \beta \frac{\left(\frac{b_1^{NC}}{\bar{y}} \right)^2}{\left(\frac{b_2^{NC}}{\bar{y}} \right)^2} < 0.$$

But at (b_1^{NC}, b_2^{NC}) , the no-commitment FOC (33) requires

$$x^\dagger(b_1^{NC}) \left[1 - \beta \frac{\left(\frac{b_1^{NC}}{\bar{y}} \right)^2}{\left(\frac{b_2^{NC}}{\bar{y}} \right)^2} \right] - \frac{1}{2} \left(\frac{b_1^{NC}}{\bar{y}} \right) \frac{4\beta \left(\frac{b_2^{NC}}{\bar{y}} \right) + 1}{4\beta \left(\frac{b_2^{NC}}{\bar{y}} \right) + 3} = 0.$$

Since $x^\dagger(b_1^{NC}) > 0$ and the second term is nonnegative, the bracketed expression must be nonnegative, contradicting the inequality above. Hence

$$\frac{b_2^{NC}}{b_1^{NC}} \geq \sqrt{\beta},$$

which in turn implies

$$\frac{b_2^{NC}}{b_1^{NC} + b_2^{NC}} \geq \frac{\sqrt{\beta}}{1 + \sqrt{\beta}} = \frac{b_2^C}{b_1^C + b_2^C}.$$

■

A.3 Commitment Problem

Under commitment, the period-1 government chooses both b_1 and b_2 , taking into account their joint effect on prices and default risk.

Using $B = b_1 + b_2$, $x = 1 - B/\bar{y}$, and $q(B) = \sqrt{x}$, period-1 and period-2 consumptions are $c_1 = q(B)b_1$ and $c_2 = q(B)b_2$, and the lifetime objective is

$$V_1^C(b_1, b_2) = -\frac{1}{q(B)b_1} - \frac{\beta}{q(B)b_2} + \beta^2 \mathbb{E}[u(c_3) \mid B], \quad (34)$$

with $\mathbb{E}[u(c_3) \mid B]$ given by (4).

Since $x = (\bar{y} - B)/\bar{y}$ and $q = \sqrt{x}$,

$$q'(B) = \frac{dq}{dB} = \frac{dq}{dx} \frac{dx}{dB} = \frac{1}{2\sqrt{x}} \left(-\frac{1}{\bar{y}} \right) = -\frac{1}{2\bar{y}\sqrt{x}},$$

and

$$\frac{d}{dB} \mathbb{E}[u(c_3) \mid B] = \frac{d}{dB} \left[\frac{1}{\bar{y}} \left(1 - 2x^{-1/2} \right) \right] = -\frac{1}{\bar{y}^2 x^{3/2}}.$$

Differentiating (34) with respect to b_1 and b_2 , and using $\partial B/\partial b_1 = \partial B/\partial b_2 = 1$, we obtain:

FOC with respect to b_1 .

$$\frac{\partial V_1^C}{\partial b_1} = \frac{q + b_1 q'}{q^2 b_1^2} + \beta \frac{q'}{q^2 b_2} + \beta^2 \frac{d}{dB} \mathbb{E}[u(c_3) \mid B] = 0. \quad (35)$$

FOC with respect to b_2 .

$$\frac{\partial V_1^C}{\partial b_2} = \frac{q'}{q^2 b_1} + \beta \frac{q + b_2 q'}{q^2 b_2^2} + \beta^2 \frac{d}{dB} \mathbb{E}[u(c_3) \mid B] = 0. \quad (36)$$

Subtracting (36) from (35) eliminates the common term $\beta^2 \frac{d}{dB} \mathbb{E}[u(c_3) \mid B]$:

$$\begin{aligned} 0 &= \frac{q + b_1 q'}{q^2 b_1^2} + \beta \frac{q'}{q^2 b_2} - \frac{q'}{q^2 b_1} - \beta \frac{q + b_2 q'}{q^2 b_2^2} \\ &= \frac{qb_2^2 - \beta qb_1^2}{q^2 b_1^2 b_2^2}, \end{aligned}$$

which implies

$$\frac{b_2^C}{b_1^C} = \sqrt{\beta}. \quad (37)$$

This is the maturity ratio reported in the main text.

Using $b_2^C = \sqrt{\beta} b_1^C$ and $B^C = b_1^C + b_2^C$, we have

$$x^C = 1 - \frac{B^C}{\bar{y}} = 1 - \frac{b_1^C}{\bar{y}}(1 + \sqrt{\beta}), \quad q^C = \sqrt{x^C}.$$

Substituting these expressions into (35) (or equivalently (36)) and simplifying yields a quadratic in $b_1^C/\bar{y}$:

$$\beta^2 \left(\frac{b_1^C}{\bar{y}} \right)^2 + \frac{3(1 + \sqrt{\beta})}{2} \left(\frac{b_1^C}{\bar{y}} \right) - 1 = 0.$$

The positive root is

$$\frac{b_1^C}{\bar{y}} = \frac{-\frac{3(1+\sqrt{\beta})}{2} + \sqrt{4\beta^2 + \frac{9}{4}(1 + \beta + 2\sqrt{\beta})}}{2\beta^2}, \quad (38)$$

and, by (37),

$$\frac{b_2^C}{\bar{y}} = \frac{-\frac{3(1+\sqrt{\beta})}{2} + \sqrt{4\beta^2 + \frac{9}{4}(1 + \beta + 2\sqrt{\beta})}}{2\beta^{3/2}}. \quad (39)$$

These are the closed-form expressions reported in the main text and used to construct the commitment benchmark.

A.4 Intermediate Commitment with a Debt Ceiling

We now derive the key objects for the intermediate-commitment case with a debt ceiling $\bar{b}$ and enforcement cost ϕ .

Time-2 Problem and the Value Gap

At $t = 2$, given inherited b_1 and a ceiling $\bar{b}$, the government chooses b_2 to maximize

$$V_2(b_1, \bar{b}) = \max_{b_2} \left\{ u(c_2) - \phi \mathbf{1}_{\{b_2 > \bar{b}\}} + \beta \mathbb{E}[u(c_3) \mid B] \right\},$$

with $c_2 = q(B)b_2$, $B = b_1 + b_2$, $x = 1 - B/\bar{y}$ and $q(B) = \sqrt{x}$. Define the continuation value excluding the penalty:

$$f_0(b_2 \mid b_1) = u(c_2) + \beta \mathbb{E}[u(c_3) \mid B] = -\frac{1}{b_2 \sqrt{x}} + \frac{\beta}{\bar{y}} \left(1 - 2x^{-1/2} \right), \quad x = 1 - \frac{b_1 + b_2}{\bar{y}}. \quad (40)$$

When $\phi = 0$, the FOC coincides with the no-commitment case and yields the interior best response $b_2^\dagger(b_1)$ in (27).

With the ceiling and enforcement cost, the period-2 policy is

$$b_2(b_1, \bar{b}) = \begin{cases} b_2^\dagger(b_1), & \text{if } f_0(b_2^\dagger(b_1) \mid b_1) - \phi > f_0(\bar{b} \mid b_1), \\ \bar{b}, & \text{otherwise.} \end{cases}$$

For given b_1 , a ceiling $\bar{b}$ is *enforceable* if the period-2 incentive constraint

$$f_0(b_2^\dagger(b_1) \mid b_1) - \phi \leq f_0(\bar{b} \mid b_1)$$

holds. Among enforceable ceilings, the period-1 government chooses the tightest one. This implies that, whenever the ceiling is used (i.e., $b_2(b_1, \bar{b}) = \bar{b}$ in equilibrium), it lies on the indifference locus

$$f_0(b_2^\dagger(b_1) \mid b_1) - \phi = f_0(\bar{b} \mid b_1), \quad (41)$$

which determines $\bar{b}$ as a function of b_1 .

To sustain the commitment allocation (b_1^C, b_2^C) , the planner sets $\bar{b} = b_2^C$ and chooses ϕ so

that the period-2 government is indifferent between deviating to $b_2^\dagger(b_1^C)$ and respecting the ceiling:

$$f_0(b_2^\dagger(b_1^C) \mid b_1^C) - \phi = f_0(b_2^C \mid b_1^C).$$

The smallest enforcement level that satisfies this is the *value gap*

$$\phi_{\min} = f_0\left(b_2^\dagger(b_1^C) \mid b_1^C\right) - f_0(b_2^C \mid b_1^C) > 0, \quad (42)$$

which coincides with equation (14) in the main text.

Derivatives of the Continuation Value

For notational convenience, define

$$x(b_1, b_2) = 1 - \frac{b_1 + b_2}{\bar{y}}, \quad s(b_1, b_2) = \sqrt{x(b_1, b_2)}.$$

Then the continuation value (40) can be written as

$$V(b_1, b_2) = -\frac{1}{b_2 s(b_1, b_2)} + \frac{\beta}{\bar{y}} \left(1 - \frac{2}{s(b_1, b_2)}\right). \quad (43)$$

We will use this representation to derive the generalized Euler equation under intermediate commitment.

Using $x = 1 - (b_1 + b_2)/\bar{y}$ and $\partial x / \partial b_j = -1/\bar{y}$ for $j = 1, 2$, we obtain

$$\frac{\partial s}{\partial b_j} = \frac{1}{2} x^{-1/2} \frac{\partial x}{\partial b_j} = -\frac{1}{2\bar{y}} x^{-1/2}, \quad \frac{\partial}{\partial b_j} \left(\frac{1}{s}\right) = \frac{1}{2\bar{y}} x^{-3/2}, \quad j = 1, 2.$$

Differentiating (43) gives:

Derivative with respect to b_1 .

$$\begin{aligned} \frac{\partial V}{\partial b_1}(b_1, b_2) &= -\frac{1}{b_2} \frac{\partial}{\partial b_1} \left(\frac{1}{s}\right) - \frac{2\beta}{\bar{y}} \frac{\partial}{\partial b_1} \left(\frac{1}{s}\right) \\ &= -\left(\frac{1}{2\bar{y} b_2} + \frac{\beta}{\bar{y}^2}\right) x(b_1, b_2)^{-3/2}. \end{aligned} \quad (44)$$

Derivative with respect to b_2 .

$$\begin{aligned}\frac{\partial V}{\partial b_2}(b_1, b_2) &= \frac{1}{b_2^2 s} - \frac{1}{b_2} \frac{\partial}{\partial b_2} \left(\frac{1}{s} \right) - \frac{2\beta}{\bar{y}} \frac{\partial}{\partial b_2} \left(\frac{1}{s} \right) \\ &= \frac{1}{b_2^2 \sqrt{x(b_1, b_2)}} - \left(\frac{1}{2\bar{y} b_2} + \frac{\beta}{\bar{y}^2} \right) x(b_1, b_2)^{-3/2}.\end{aligned}\tag{45}$$

Envelope Result for the Disciplined Continuation Value

For each b_1 , denote by $b_2^*(b_1)$ the disciplined policy that respects the ceiling and solves the indifference condition (41), and define the associated continuation value

$$V^*(b_1) = V(b_1, b_2^*(b_1)).$$

For comparison, denote by $b_2^\dagger(b_1)$ the unconstrained best response from (27). A standard implicit-function argument shows that

$$\frac{dV^*}{db_1}(b_1) = \frac{\partial V}{\partial b_1}(b_1, b_2^\dagger(b_1)) + \frac{\partial V}{\partial b_2}(b_1, b_2^\dagger(b_1)) \frac{db_2^\dagger}{db_1}(b_1),\tag{46}$$

so dV^*/db_1 depends only on the unconstrained threat policy $b_2^\dagger(b_1)$ and its derivative. Substituting (44), (45), and (29) into (46) yields an explicit expression for dV^*/db_1 in terms of $b_2^\dagger$, $x(b_1, b_2^\dagger)$, and the parameters $(\beta, \bar{y})$.

Time-1 Problem and Proof of Proposition 2

At time 1, with a ceiling policy that induces $b_2^*(b_1)$, the government chooses b_1 to maximize

$$V_1(b_1) = u(c_1) + \beta V^*(b_1) = -\frac{1}{\sqrt{x^*(b_1)} b_1} + \beta V^*(b_1),$$

where

$$x^*(b_1) = 1 - \frac{b_1 + b_2^*(b_1)}{\bar{y}}.$$

The FOC is

$$0 = -\frac{1 + b_2'^*(b_1)}{2\bar{y}} \frac{(x^*(b_1))^{-3/2}}{b_1} + \frac{(x^*(b_1))^{-1/2}}{b_1^2} + \beta \frac{dV^*}{db_1}(b_1),\tag{47}$$

where dV^*/db_1 is given by (46). Rewriting (47) in terms of $B = b_1 + b_2^*(b_1)$ and $x^* = 1 - B/\bar{y}$ yields the generalized Euler equation (15) in the main text.

We are now in a position to prove Proposition 2.

Proof of Proposition 2. Part (1) follows immediately from the fact that when $\phi = 0$ the penalty term in $V_2(b_1, \bar{b})$ is inactive, so the period-2 best response is $b_2^\dagger(b_1)$ from (27). The period-1 FOC then coincides with the no-commitment condition in the main text (equivalently (33) here), yielding (b_1^{NC}, b_2^{NC}) .

For part (2), fix $0 < \phi < \phi_{\min}$. For each b_1 , the ceiling is enforceable only if the period-2 incentive constraint

$$f_0(b_2^\dagger(b_1) \mid b_1) - \phi \leq f_0(\bar{b} \mid b_1)$$

is satisfied. For each b_1 , the period-1 government prefers the tightest ceiling that the period-2 government is willing to respect. Hence the optimal enforceable ceiling lies on the indifference condition (41). At this ceiling, $b_2(b_1, \bar{b}) = \bar{b} = b_2^*(b_1)$, and substituting $B = b_1 + b_2^*(b_1)$ into the period-1 problem yields the generalized Euler equation (15). Comparing with the commitment and no-commitment benchmarks, and using the fact that the ceiling partially disciplines the period-2 overborrowing motive but does not eliminate it, implies $b_1^* \in (b_1^C, b_1^{NC})$ and $b_2^* \in (b_2^C, b_2^{NC})$.

Part (3) follows from the definition of $\phi_{\min}$ in (42). For $\phi = \phi_{\min}$, the ceiling $\bar{b} = b_2^C$ exactly satisfies the incentive constraint at $b_1 = b_1^C$, making the commitment allocation (b_1^C, b_2^C) self-enforcing at $t = 2$. For any $\phi > \phi_{\min}$, deviating to $b_2^\dagger(b_1^C)$ is strictly worse for the period-2 government, so the same ceiling remains enforceable and (b_1^C, b_2^C) solves the period-1 problem. Thus, for all $\phi \geq \phi_{\min}$ the unique sustainable allocation coincides with the commitment benchmark. ■

Intermediate commitment with a quadratic cost ($\zeta = 2$). This case yields a tractable first-order condition for the period-2 government and therefore provides a useful contrast with the environments $\zeta = 0$ and $\zeta = 1$. Numerical results for both $\zeta = 1$ and $\zeta = 2$ are presented in Figure 3, which illustrate the different ways in which ceilings shape borrowing incentives.

Given $(b_1, \bar{b})$, the period-2 government chooses b_2 to maximize

$$V_2(b_1, \bar{b}) = \max_{b_2} \left\{ u(c_2) - \phi(b_2 - \bar{b})^2 \mathbf{1}_{\{b_2 > \bar{b}\}} + \beta \mathbb{E}[u(c_3) \mid B] \right\},$$

with $c_2 = q(B)b_2$ and $B = b_1 + b_2$. Define

$$f_0(b_2 \mid b_1) = u(c_2) + \beta \mathbb{E}[u(c_3) \mid B].$$

For $b_2 \leq \bar{b}$, the penalty is inactive and the first-order condition is identical to the no-commitment

case (9). For an interior optimum with $b_2 > \bar{b}$, differentiating the objective yields

$$\frac{\partial}{\partial b_2} f_0(b_2 | b_1) - 2\phi(b_2 - \bar{b}) = 0, \quad b_2 > \bar{b}, \quad (48)$$

where

$$\frac{\partial}{\partial b_2} f_0(b_2 | b_1) = u'(c_2)(q(B) + b_2 q'(B)) + \beta \frac{\partial}{\partial B} \mathbb{E}[u(c_3) | B].$$

Relative to the unconstrained best response $b_2^\dagger(b_1)$ in (10), the quadratic penalty introduces a smooth marginal cost $2\phi(b_2 - \bar{b})$ that grows linearly with the violation. Let $b_2^Q(b_1, \bar{b})$ denote the policy function solving the first-order condition (48). Because the quadratic penalty is differentiable at $b_2 = \bar{b}$, the borrowing response is smooth whenever the ceiling is exceeded, and violations can be characterized by the interior first-order condition below.

Anticipating $b_2^Q(b_1, \bar{b})$, the period-1 government chooses $(b_1, \bar{b})$ to maximize

$$V_1^Q(b_1, \bar{b}) = u(c_1) + \beta \left(u(c_2^Q) - \phi(b_2^Q - \bar{b})^2 \mathbf{1}_{\{b_2^Q > \bar{b}\}} \right) + \beta^2 \mathbb{E}[u(c_3) | B^Q],$$

with $c_1 = q(B^Q)b_1$, $c_2^Q = q(B^Q)b_2^Q(b_1, \bar{b})$, and $B^Q = b_1 + b_2^Q(b_1, \bar{b})$. Differentiating with respect to b_1 yields the generalized Euler equation

$$\begin{aligned} 0 = & u'(c_1) \left(q(B^Q) + b_1 q'(B^Q)(1 + b_2^{Q'}(b_1, \bar{b})) \right) \\ & + \beta \left[u'(c_2^Q) \left(q(B^Q) b_2^{Q'} + q'(B^Q) b_2^Q (1 + b_2^{Q'}) \right) + \beta \frac{\partial}{\partial B} \mathbb{E}[u(c_3) | B^Q] (1 + b_2^{Q'}) \right], \end{aligned} \quad (49)$$

where $b_2^{Q'}$ denotes $\partial b_2^Q / \partial b_1$. Equation (49) has the identical structure to the no-commitment Euler equation, except that the policy-response elasticity $b_2^{\dagger'}(b_1)$ is replaced by the ceiling-induced response $b_2^{Q'}(b_1, \bar{b})$. As with the fixed-penalty case, the ceiling reduces the sensitivity of period-2 issuance to inherited debt, thereby partially restoring the discipline present under commitment.

To characterize the optimal ceiling, differentiate $V_1^Q(b_1, \bar{b})$ with respect to $\bar{b}$. Because b_2^Q solves the period-2 problem, the envelope theorem applies to V_2 , and the derivative decomposes into two terms:

$$\frac{\partial V_1^Q}{\partial \bar{b}} = u'(c_1) q'(B^Q) b_1 \frac{\partial b_2^Q}{\partial \bar{b}} + 2\beta \phi(b_2^Q - \bar{b}) \mathbf{1}_{\{b_2^Q > \bar{b}\}}.$$

The second term captures the direct effect of relaxing the ceiling on the penalty paid at $t = 2$, while the first term reflects how the ceiling influences period-1 resources through its effect

on the equilibrium borrowing b_2^Q and thus the bond price $q(B)$. In order to characterize the region in which the quadratic penalty is active, we focus on the region where $b_2^Q > \bar{b}$, so the indicator $\mathbf{1}_{\{b_2^Q > \bar{b}\}}$ equals one throughout and can be dropped. Setting the derivative equal to zero yields the optimality condition

$$u'(c_1) q'(B^Q) b_1 \frac{\partial b_2^Q}{\partial \bar{b}} + 2\beta\phi (b_2^Q - \bar{b}) = 0. \quad (50)$$

Equation (50) shows that, unlike the fixed-cost case $\zeta = 0$, a quadratic penalty does *not* imply $b_2^Q = \bar{b}$ in equilibrium. Relaxing the ceiling reduces the penalty but also increases b_2^Q , which lowers the bond price $q(B)$ and therefore depresses c_1 . The optimal ceiling balances these two forces, generating an interior gap $b_2^Q - \bar{b} > 0$. Figure 3 shows that under a quadratic penalty ($\zeta = 2$), the gap between b_2 and $\bar{b}_2$ shrinks as ϕ increases, but vanishes only asymptotically as $\phi \rightarrow \infty$, at which point the allocation converges to the commitment benchmark. By contrast, under a linear penalty ($\zeta = 1$), commitment is restored at a finite threshold $\phi_{\min}$, similarly to the $\zeta = 0$ case.

A.5 Proof of Proposition 3

Fix $(b_1, \bar{b})$ and state $s \in \{L, H\}$. Under the fixed-cost specification, the period-2 government compares the value from respecting the ceiling,

$$f_0(\bar{b} \mid b_1, y_s),$$

with the value from choosing the unconstrained optimum and paying the fixed cost,

$$f_0(b_2^\dagger(b_1, y_s) \mid b_1, y_s) - \phi.$$

Therefore the government breaches the ceiling in state s if and only if

$$f_0(b_2^\dagger(b_1, y_s) \mid b_1, y_s) - \phi > f_0(\bar{b} \mid b_1, y_s),$$

or equivalently,

$$\Delta_s(b_1, \bar{b}) > \phi.$$

If $\phi \geq \Delta_L(b_1, \bar{b})$, then since $\Delta_L(b_1, \bar{b}) > \Delta_H(b_1, \bar{b})$, we also have $\phi \geq \Delta_H(b_1, \bar{b})$. Hence the ceiling is respected in both states.

If

$$\Delta_H(b_1, \bar{b}) \leq \phi < \Delta_L(b_1, \bar{b}),$$

then the ceiling is respected in the high-output state but breached in the low-output state.

Finally, if $\phi < \Delta_H(b_1, \bar{b})$, then since $\Delta_L(b_1, \bar{b}) > \Delta_H(b_1, \bar{b})$, the fixed cost is below the breach value in both states. Hence the ceiling is breached in both states. This proves the result. ■

B Computational Strategy

This appendix describes how we compute the Markov-perfect equilibrium of the infinite-horizon model with long-term debt and endogenous debt-ceiling announcements.

B.1 State Space, Grids, and Interpolation

The government's state at the beginning of a period is

$$(y, B, \bar{B}),$$

where y is the current endowment realization, B is the stock of outstanding long-term debt carried into the period, and $\bar{B}$ is the inherited debt ceiling announced in the previous period. The exogenous process for y follows the AR(1) specification described in Section 3, discretized using a finite Markov chain with N_y nodes and transition matrix P_y .

We approximate the endogenous state variables on finite grids:

- A grid $\mathcal{B} = \{B_1, \dots, B_{N_B}\}$ for outstanding long-term debt.
- A grid $\bar{\mathcal{B}} = \{\bar{B}_1, \dots, \bar{B}_{N_{\bar{B}}}\}$ for the announced debt ceiling.

Bounds $[B_{\min}, B_{\max}]$ and $[\bar{B}_{\min}, \bar{B}_{\max}]$ are chosen to be wide enough that the equilibrium support of debt and ceilings lies in the interior of each grid. We use evenly spaced points for both B and $\bar{B}$ in the baseline implementation.

For expectations and bond pricing, we repeatedly evaluate objects such as value functions, default probabilities, and policy probabilities at off-grid points. To do so we construct multidimensional interpolants (Matlab's `griddedInterpolant`) on the tensor product grid

$$\mathcal{S} = \mathcal{B} \times \bar{\mathcal{B}} \times \mathcal{Y},$$

and use multilinear interpolation in $(B, \bar{B}, y)$ whenever necessary.

B.2 Extreme-Value Formulation and Policy Probabilities

We follow [Dvorkin et al. \(2021\)](#) and solve the government's problem in a stochastic choice environment with additive Type I extreme-value shocks. For each state $(y, B, \bar{B})$ the government has two types of actions:

1. Default: exit financial markets, consume $\varrho(y)$, and transition into autarky according to (23).
2. Repay and choose next period's debt and ceiling $(B', \bar{B}')$, subject to the implementability constraint (20) and the ceiling cost $\Phi(B', \bar{B})$.

Let $v^D(y, B, \bar{B})$ denote the continuation value from default, and let

$$v^R(y, B, \bar{B}; B', \bar{B}')$$

denote the value from repaying and choosing $(B', \bar{B}')$ optimally in the future.

We assume that:

- The default option is subject to an idiosyncratic taste shock ε^D .
- Each repayment alternative $(B', \bar{B}')$ is subject to a shock $\varepsilon^R(B', \bar{B}')$.

Shocks are Type I extreme value, with scale parameter $v > 0$, and we allow for correlation among repayment alternatives through a correlation parameter $p \in (0, 1]$, which nests the standard logit when $p = 1$. Under this specification, the probability of choosing a particular repayment action $(B', \bar{B}')$ conditional on repaying is

$$G(B', \bar{B}' \mid y, B, \bar{B}) = \frac{\exp\left(\frac{v^R(y, B, \bar{B}; B', \bar{B}')}{pv}\right)}{\sum_{(B'', \bar{B}'')} \exp\left(\frac{v^R(y, B, \bar{B}; B'', \bar{B}'')}{pv}\right)},$$

and the probability of default is

$$D(y, B, \bar{B}) = \frac{\exp\left(\frac{v^D(y, B, \bar{B})}{v}\right)}{\exp\left(\frac{v^D(y, B, \bar{B})}{v}\right) + \left[\sum_{(B'', \bar{B}'')} \exp\left(\frac{v^R(y, B, \bar{B}; B'', \bar{B}'')}{pv}\right)\right]^p}.$$

In the implementation we work with “adjusted” value objects that factor out common components of v^R and v^D and compute G and D using vectorized expressions based on these formulas.

B.3 Joint Iteration on Values and Bond Prices

Equilibrium is characterized by a fixed point in the space of (i) government value functions, (ii) default and policy probabilities, and (iii) the bond-price function $q(y, B', \bar{B}')$. We solve for this fixed point by iterating on the joint mapping induced by the government’s Bellman equation and the lenders’ zero-profit condition.

Initialization. We start from an initial guess for:

- The repayment value kernel $v_0^R(y, B, \bar{B}; B', \bar{B}')$ (equivalently, the object denoted v in the code).
- The default value $v_0^D(y, B, \bar{B})$.
- The bond-price function $q_0(y, B', \bar{B}')$.

The initial conditions are either taken from the no-ceiling benchmark or from a previously solved nearby parameterization to accelerate convergence.

Step 1: Probabilities G and D . Given current value objects (v_t^R, v_t^D) , we compute:

- The conditional repayment choice probabilities $G_t(B', \bar{B}' \mid y, B, \bar{B})$ as in the logit formula above.
- The default probability $D_t(y, B, \bar{B})$.

Both objects are stored on the tensor-product grid and represented by `griddedInterpolant` objects for efficient evaluation.

Step 2: Expected continuation values. Using G_t and D_t , we compute expected continuation values. For each state $(y, B, \bar{B})$ we form

$$\mathbb{E}_y[V(y', B', \bar{B}')] = \sum_{y'} P_y(y' \mid y) [D_t(y', B', \bar{B}') v_t^D(y', B', \bar{B}') + (1 - D_t(y', B', \bar{B}')) \tilde{v}_t^R(y', B', \bar{B}')] ,$$

where $\tilde{v}_t^R$ is the expected value under the repayment-choice distribution G_t in the following period.

Step 3: Updating the value kernels. We then update the repayment and default values using the Bellman equations:

$$\begin{aligned} v_{t+1}^R(y, B, \bar{B}; B', \bar{B}') &= u(c(y, B, \bar{B}; B', \bar{B}')) + \beta \mathbb{E}_y[V_t(y', B', \bar{B}')], \\ v_{t+1}^D(y, B, \bar{B}) &= u(\varrho(y)) + \beta \mathbb{E}_y[\theta V_t(y', 0, \bar{B}_{\max}) + (1 - \theta) v_t^D(y')], \end{aligned}$$

where consumption $c(\cdot)$ satisfies the implementability constraint with the chosen ceiling cost,

$$c + (\delta + (1 - \delta)z)B = y + q_t(y, B', \bar{B}')(B' - (1 - \delta)B) - \Phi(B', \bar{B}) - \iota(B', B).$$

To ensure numerical stability we apply a relaxation step:

$$v_{t+1}^R \leftarrow \lambda v_{t+1}^R + (1 - \lambda) v_t^R, \quad v_{t+1}^D \leftarrow \lambda v_{t+1}^D + (1 - \lambda) v_t^D,$$

with $\lambda \in (0, 1)$.

Step 4: Updating the bond-price function. Given default probabilities $D_t(y', B', \bar{B}')$ and the repayment-policy distribution G_t , we impose the lenders' zero-profit condition to obtain a new price function. For each $(y, B', \bar{B}')$,

$$q_{t+1}(y, B', \bar{B}') = \frac{1}{1 + r} \mathbb{E}_{y'} \left[(1 - D_t(y', B', \bar{B}')) (\delta + (1 - \delta)z + (1 - \delta) \mathbb{E}_{(B'', \bar{B}'')|G_t} [q_t(y', B'', \bar{B}'')]) \right],$$

where the inner expectation integrates over next-period debt and ceiling choices using G_t .¹⁴

Step 5: Convergence. Steps 1–4 are iterated until the sup-norm of value-function differences is below a small tolerance,

$$\max \{ \|v_{t+1}^R - v_t^R\|_\infty, \|v_{t+1}^D - v_t^D\|_\infty \} < 10^{-5},$$

at which point we treat (v^R, v^D, G, D, q) as an equilibrium fixed point for the given value of the debt-ceiling penalty parameter ϕ and cost curvature ζ .

¹⁴In the code this is implemented by looping over future grid points $(B'', \bar{B}'')$, accumulating the continuation price, and then averaging with respect to both the income transition matrix and the policy probabilities.

B.4 Debt-Ceiling Costs and Variants

The cost of exceeding the debt ceiling is given by

$$\Phi(B', \bar{B}) = \begin{cases} 0, & B' \leq \bar{B}, \\ \phi(B' - \bar{B})^\zeta, & B' > \bar{B}, \end{cases}$$

and we consider two cases in the quantitative analysis:

- **Fixed-cost case** ($\zeta = 0$): any breach of the announced ceiling triggers a discrete penalty ϕ , independent of the size of the violation.
- **Quadratic-cost case** ($\zeta = 2$): larger violations of the ceiling are disproportionately more costly, $\Phi \propto (B' - \bar{B})^2$.

For each specification we solve the equilibrium using the joint iteration described above. The numerical algorithm is identical across cases; only the evaluation of $\Phi(B', \bar{B})$ changes.

B.5 Simulation and Welfare Computation

Once the equilibrium objects (G, D, q) have been computed for a given ϕ , we simulate the model to obtain model-implied moments and welfare.

Simulation. We simulate long histories of the Markov state $(y_t, B_t, \bar{B}_t)$ using:

1. Draw a sequence of shocks $\{y_t\}$ from the Markov chain P_y .
2. At each t , compute default $D(y_t, B_t, \bar{B}_t)$ and repayment probabilities $G(B_{t+1}, \bar{B}_{t+1} \mid y_t, B_t, \bar{B}_t)$.
3. Draw default vs repayment using D , and conditional on repayment, draw $(B_{t+1}, \bar{B}_{t+1})$ using G .

We discard an initial burn-in and use the remainder to compute the distribution of debt, default frequencies, average and volatility of spreads, and the behavior of announced ceilings.

Welfare. To evaluate the welfare effects of introducing debt-ceiling frictions, we compute expected lifetime utility under the equilibrium government policies associated with each value

of ϕ . As in Section 3.6, welfare is evaluated by a patient social planner with discount factor

$$\beta^{SP} = \frac{1}{1+r}.$$

Let $V^{SP}(y_0, B_0 \mid \phi)$ denote the planner's evaluation of welfare under the Markov-perfect government policies associated with ϕ :

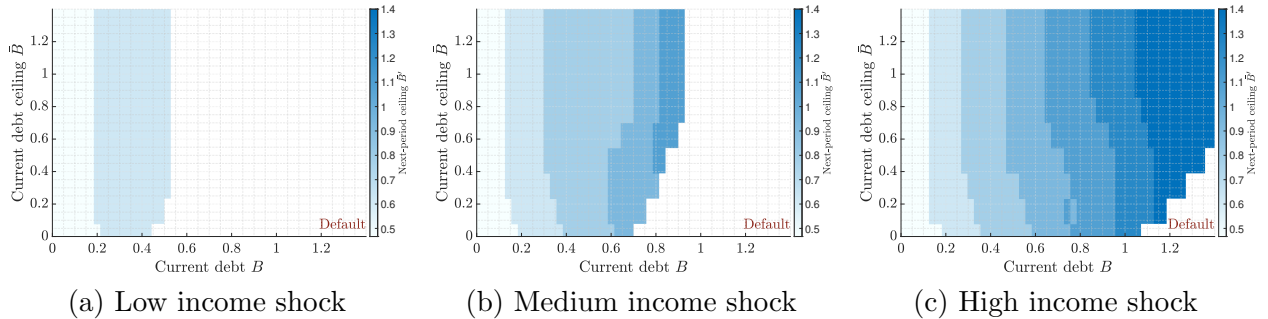
$$V^{SP}(y_0, B_0 \mid \phi) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} (\beta^{SP})^t u(c_t) \right].$$

The expectation is taken over simulated histories generated by the equilibrium policy functions. We report consumption-equivalent welfare gains relative to the benchmark without debt-ceiling costs ($\phi = 0$), defined as the constant percentage adjustment in consumption that makes the planner indifferent between the two environments.

In all cases the algorithm converges rapidly and produces smooth policy functions and bond-price schedules, thanks to the extreme-value formulation and the use of interpolation on a moderate-sized multidimensional grid.

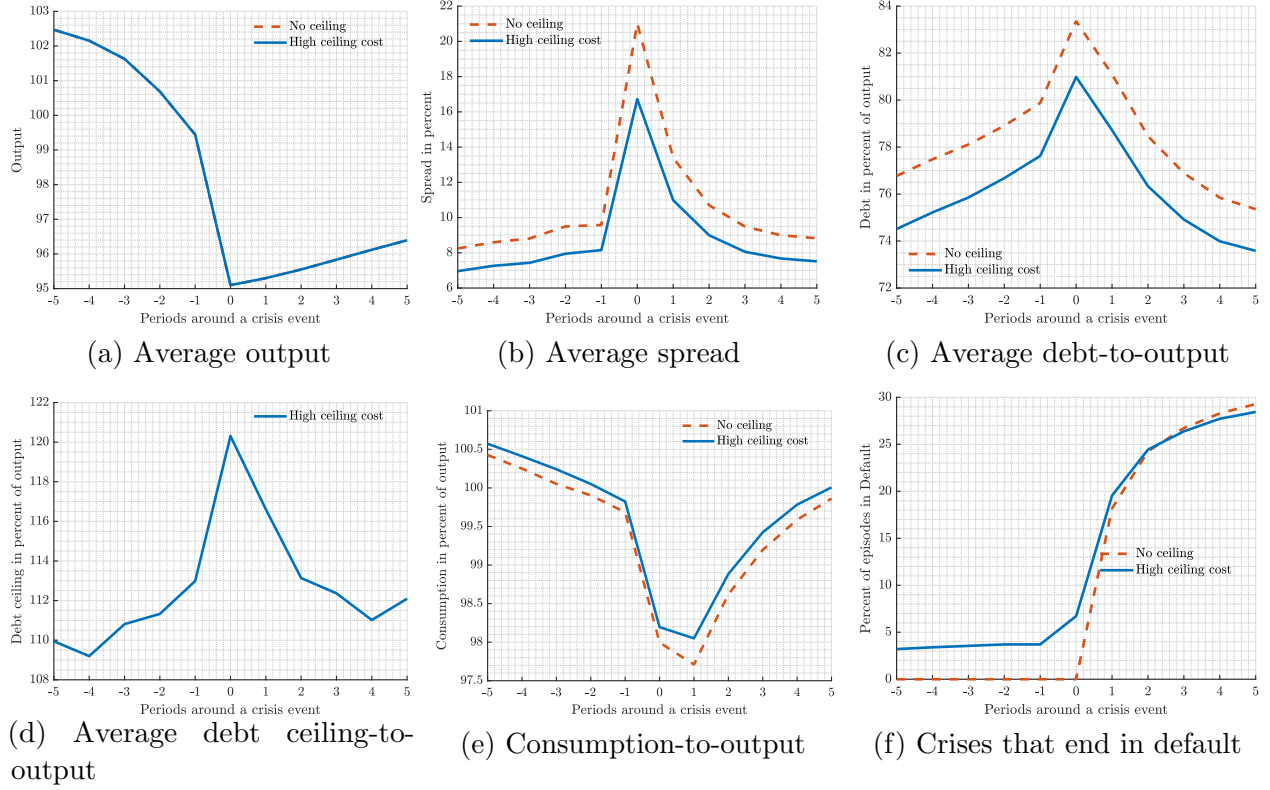
C Additional Figures and Tables

Figure 11: Debt Ceiling Policy



Note: The figure reports the debt-ceiling policy as a function of current debt and the inherited ceiling in the quadratic-cost model with $\phi = .4$. Panel **a** corresponds to the low-income realization, Panel **b** to the medium-income realization, and Panel **c** to the high-income realization.

Figure 12: Debt Ceilings During Sovereign Crises



Note: The figure reports an event analysis of a fiscal crisis in the model without a debt ceiling. A fiscal crisis is defined as an episode in which the spread increases by more than two standard deviations above its mean. We exclude any event in which this economy was in default five quarters before the crisis. We then evaluate the model with a fixed enforcement cost ($\zeta = 0$) under the exact same sequence of shocks. Yellow lines represent the model without a debt ceiling. Blue lines represent the model with a debt ceiling and the enforcement cost set to $\phi = .04$. The spread, debt, debt-ceiling, and consumption paths are computed only for events without default in either model. Panel **a** plots an average of the common output shocks. Panel **b** plots the average annualized sovereign spread. Panel **c** plots the average debt-to-output ratio. Panel **d** plots the average debt ceiling-to-output ratio. Panel **e** plots the average consumption-to-output ratio. Panel **f** plots the percentage of events that end in sovereign default around the crisis. By construction, episodes with a default before the crisis in the model without a ceiling are not considered.

Table 2: Regression Results for Monthly EMBIG

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lagged dummies	Full model dummies	Lagged haircuts	Full haircuts	Dummies + haircuts	+ rating	Fundamentals	Full model
Fiscal rule	-42.168* (23.690)	-59.280*** (20.931)	-44.871* (23.436)	-60.552*** (20.296)	-34.343 (23.241)	-34.800* (21.016)	-42.239* (22.292)	-52.712** (20.919)
Restructuring dummy, 1 year lag	193.362*** (62.135)	116.593* (65.816)			217.225** (107.542)	222.122** (113.010)	97.124 (109.145)	91.950 (118.450)
Restructuring dummy, 2 year lag	89.724* (50.654)	100.703** (49.986)			79.590 (96.852)	79.192 (99.556)	34.454 (95.717)	87.405 (96.803)
Restructuring dummy, 3 year lag	29.323 (43.674)	50.934 (38.757)			-95.417 (68.717)	-96.716 (59.653)	-87.042 (64.259)	-20.725 (62.423)
Restructuring dummy, 4 and 5 year lag	13.717 (33.621)	50.128* (28.860)			-149.403** (73.299)	-150.527** (60.360)	-116.785* (66.517)	-82.993 (58.449)
Restructuring dummy, 6 and 7 year lag	-3.562 (31.936)	17.732 (26.701)			-169.371*** (62.548)	-169.795*** (50.134)	-172.215*** (57.650)	-148.988*** (47.820)
US low-grade corporate yield	56.243*** (4.205)	55.155*** (4.122)	55.996*** (4.200)	54.983*** (4.120)	56.090*** (4.210)	56.099*** (4.151)	54.961*** (4.167)	55.100*** (4.128)
Residuals		-42.801*** (5.256)		-42.323*** (5.183)		-58.536*** (6.215)		-41.427*** (5.266)
Public debt to GNI		3.984*** (1.232)		4.060*** (1.213)			4.823*** (1.089)	3.857*** (1.229)
GDP real growth		-9.945*** (2.558)		-10.659*** (2.566)			-13.559*** (3.143)	-10.367*** (2.634)
Reserves to imports		-1.573*** (0.452)		-1.613*** (0.452)				-1.585*** (0.453)
Inflation		0.265** (0.123)		0.214* (0.124)				0.198 (0.121)
Budget balance to GDP		-12.366*** (3.867)		-12.494*** (3.832)				-12.219*** (3.933)
Current account to GDP		-6.228*** (1.637)		-5.789*** (1.591)				-5.826*** (1.647)
Political Risk Index (ICRG)		-12.593*** (1.665)		-12.765*** (1.683)			-14.084*** (1.922)	-12.645*** (1.674)
Haircut (SZ), 1 year lag			3.414*** (1.315)	2.377** (0.981)	-0.914 (1.994)	-0.902 (1.863)	-0.117 (2.009)	0.578 (1.986)
Haircut (SZ), 2 year lag			1.853* (1.006)	2.055** (0.799)	0.027 (1.885)	0.020 (1.823)	0.740 (1.843)	0.154 (1.683)
Haircut (SZ), 3 year lag			1.594 (1.015)	1.545* (0.797)	3.323** (1.679)	3.331** (1.405)	2.845** (1.427)	1.828 (1.266)
Haircut (SZ), 4 and 5 year lag			1.233* (0.688)	1.787*** (0.619)	4.058*** (1.522)	4.076*** (1.276)	3.899*** (1.274)	3.343*** (1.277)
Haircut (SZ), 6 and 7 year lag			0.875 (0.608)	1.289** (0.526)	4.159*** (1.179)	4.167*** (1.005)	4.327*** (1.075)	4.179*** (0.923)
Constant	-65.316* (36.288)	772.939*** (119.196)	-64.273* (36.640)	789.493*** (121.066)	-66.876* (36.385)	-66.584* (35.365)	836.802*** (132.951)	780.092*** (119.708)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	9194	9186	9194	9186	9194	9186	9194	9186
R-squared	0.596	0.690	0.595	0.691	0.603	0.661	0.648	0.693
Adjusted. R-squared	0.593	0.687	0.591	0.687	0.598	0.657	0.644	0.689

Notes: This table reports coefficients from unbalanced panel regressions of monthly EMBIG sovereign yield spreads—measured as the average country spread over U.S. Treasury bonds (EMBIG stripped spread, in basis points)—on fiscal rules, restructuring history, haircut measures, global risk factors, and macroeconomic fundamentals. The sample covers 57 emerging and developing economies over 1993–2019, augmenting the *Cruces and Trebesch (2013)* dataset with a fiscal-rule indicator constructed from the IMF Fiscal Rules Database (the same source underlying Figure 1) and using monthly restructuring and haircut variables following *Cruces and Trebesch (2013)* and *Arce and Fourakis (2025)*. The country sample includes Angola, Argentina, Azerbaijan, Bulgaria, Belarus, Bolivia, Brazil, Chile, China, Cameroon, Colombia, Costa Rica, Dominican Republic, Ecuador, Egypt, Ethiopia, Gabon, Ghana, Guatemala, Honduras, Croatia, Hungary, Indonesia, India, Iraq, Jamaica, Jordan, Kazakhstan, Kenya, Lebanon, Sri Lanka, Lithuania, Morocco, Mexico, Mongolia, Mozambique, Malaysia, Nigeria, Pakistan, Panama, Peru, Philippines, Papua New Guinea, Poland, Paraguay, Russia, El Salvador, Serbia, Thailand, Tunisia, Turkey, Ukraine, Uruguay, Venezuela, Vietnam, South Africa, and Zambia. Country-specific fundamentals are lagged 12 months, while the U.S. low-grade corporate yield and the ICRG Political Risk Index are lagged 1 month. All regressions include country and time fixed effects, and robust standard errors are clustered at the country level.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Table 3: Regression Results for Monthly Standard Deviation of Daily EMBIG

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lagged dummies	Full model dummies	Lagged haircuts	Full haircuts	Dummies + haircuts	Dummies + haircuts + rating	Fundamentals	Full model
Fiscal rule	-4.755*** (1.688)	-4.888*** (1.592)	-4.859*** (1.656)	-4.903*** (1.571)	-4.301** (1.670)	-4.236*** (1.616)	-4.602*** (1.652)	-4.524*** (1.601)
Restructuring dummy, 1 year lag	-3.704 (4.195)	-7.229 (4.506)			1.237 (6.395)	1.191 (6.601)	-5.056 (6.942)	-4.897 (7.171)
Restructuring dummy, 2 year lag	-2.645 (4.814)	-1.876 (5.026)			2.596 (8.843)	2.576 (9.050)	0.594 (8.901)	3.377 (9.173)
Restructuring dummy, 3 year lag	-7.135* (3.888)	-5.920 (3.888)			-6.216 (6.606)	-6.298 (6.495)	-5.523 (6.375)	-2.023 (6.487)
Restructuring dummy, 4 and 5 year lag	-3.488 (3.065)	-2.115 (2.859)			-12.187** (5.755)	-12.256** (5.330)	-10.586* (5.396)	-8.195 (5.245)
Restructuring dummy, 6 and 7 year lag	-4.078* (2.412)	-3.142 (2.342)			-14.147*** (4.502)	-14.184*** (4.240)	-14.488*** (4.274)	-13.128*** (4.236)
US low-grade corporate yield	3.885*** (0.453)	3.836*** (0.452)	3.876*** (0.453)	3.832*** (0.453)	3.877*** (0.453)	3.877*** (0.454)	3.827*** (0.451)	3.834*** (0.452)
Residuals		-1.746*** (0.369)		-1.741*** (0.365)		-2.549*** (0.433)		-1.692*** (0.371)
Public debt to GNI		0.139 (0.096)		0.142 (0.096)			0.211** (0.089)	0.130 (0.096)
GDP real growth		-0.583*** (0.191)		-0.564*** (0.192)			-0.802*** (0.217)	-0.587*** (0.195)
Reserves to imports		-0.040 (0.034)		-0.038 (0.034)				-0.040 (0.034)
Inflation		0.032*** (0.008)		0.032*** (0.008)				0.030*** (0.008)
Budget balance to GDP		-0.749*** (0.265)		-0.764*** (0.264)				-0.749*** (0.269)
Current account to GDP		-0.258** (0.116)		-0.278** (0.113)				-0.257** (0.120)
Political Risk Index (ICRG)		-0.596*** (0.143)		-0.601*** (0.142)			-0.679*** (0.153)	-0.605*** (0.144)
Haircut (SZ), 1 year lag			-0.109 (0.072)	-0.150** (0.071)	-0.156 (0.108)	-0.155 (0.105)	-0.101 (0.115)	-0.074 (0.118)
Haircut (SZ), 2 year lag			-0.084 (0.073)	-0.071 (0.074)	-0.164 (0.145)	-0.167 (0.149)	-0.127 (0.147)	-0.161 (0.149)
Haircut (SZ), 3 year lag			-0.141* (0.081)	-0.143* (0.073)	-0.041 (0.139)	-0.042 (0.135)	-0.066 (0.128)	-0.123 (0.126)
Haircut (SZ), 4 and 5 year lag			-0.014 (0.062)	-0.004 (0.056)	0.209* (0.118)	0.208* (0.109)	0.204* (0.105)	0.146 (0.104)
Haircut (SZ), 6 and 7 year lag			-0.018 (0.046)	-0.003 (0.045)	0.251*** (0.086)	0.247*** (0.085)	0.266*** (0.084)	0.249*** (0.082)
Constant	-9.527*** (3.608)	28.370*** (10.371)	-9.754*** (3.696)	28.145*** (10.386)	-9.613*** (3.626)	-9.608*** (3.637)	34.665*** (10.385)	29.052*** (10.431)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	8996	8989	8996	8989	8996	8989	8996	8989
R-squared	0.245	0.269	0.244	0.269	0.248	0.260	0.261	0.271
Adj. R-squared	0.238	0.262	0.237	0.261	0.240	0.252	0.253	0.263

Notes: This table reports coefficients from unbalanced panel regressions with robust standard errors clustered at the country level. The dependent variable is the monthly volatility of daily EMBIG returns, measured as their standard deviation. The dataset, country coverage, and variable construction follow Table 2. The analysis covers 1994–2019. Country-specific fundamentals are lagged 12 months, while the U.S. low-grade corporate yield and the ICRG Political Risk Index are lagged 1 month.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.